TM Forum Specification

Performance Thresholding User Guide

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Introduction

Performance Threshold Management API Specification

This document defines the REST API for managing performance thresholds. It includes the model definitions and all available operations necessary for creating, modifying, retrieving, and deleting performance thresholds and associated rules.

The Performance Threshold Management API enables systems to identify exceptional behavior in monitored performance indicators. It is designed as an additional layer on top of standard Performance Monitoring systems.

Overview

Performance Threshold Management focuses on identifying and reacting to abnormal values in performance metrics. Applications that generate performance indicators (PM Producing Applications) are expected to:

- Set thresholds on those indicators
- Trigger alarms when thresholds are breached
- Send alarm clear events when the values return to normal ranges

This system enhances observability and automated responses to performance degradation or unusual behavior.

Possible Actions

The API allows the following operations:

- Creating, retrieving, modifying, and deleting **threshold rules**
- Creating, retrieving, modifying, and deleting **thresholds**
- Creating, retrieving, modifying, and deleting **threshold jobs**

Purpose

The Performance Thresholding API provides a standardized interface to Service and Resource Performance Management Systems. It enables manipulation of:

- Threshold rules
- Violation rules
- Exception rules

It supports:

- Alarms and notifications when exceptions occur
- Scheduling the evaluation of thresholds and violations

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Clients

Typical API consumers (clients) include:

- Resource Performance Monitoring Applications
- Resource Performance Analysis Applications

As part of the HIP (Hybrid Infrastructure Platform), this API is exposed for use by:

- Platform capability consumers
- Platform Capability Managers for capability delivery assurance

Capabilities

The API supports the ability to define a **Threshold** as a collection of rules. These rules determine:

- When an alarm should be raised
- When an alarm should be cleared
- What severity level applies to each alarm

Each rule includes the expected fields to be set in the **Threshold Crossing Alarm (TCA)** that is triggered.

Supported Operations

The API supports:

- Creation of Performance Thresholds
- Deletion of Performance Thresholds
- Querying of existing Performance Thresholds

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Sample Use Cases

Overview

A Thresholding mechanism is a significant tool to identify exceptional behaviour at different management layers, at network level, service level and even customer level. It is assumed that a Performance Management system is deployed to collect and calculate measurements (e.g. Counters, KPIs, KQIs) that have significance to various departments of the Service Operator:

- Operations (e.g., Network uptime)
- Engineering (e.g., Packet loss rate)
- Marketing (e.g., Customer churn rate)
- Customer Care (e.g., Average call handling time)

Performance Management systems are sometimes limited to a specific domain or technology and sometimes much wider serving multiple domains, multiple technologies and multiple management layers.

In many typical scenarios, there is a need of management systems to get the information on exceptional behaviour as reflected in values of KPIs. This may be needed as a tool to identify lack of network resources (e.g. Bandwidth, CPU, transport links) to realize an insufficient Quality of Service or even bad customer experience.

As shown in the diagram below, a management system that requires the information on exceptional behaviour of KPIs is typically requesting this information from the Performance management system that collects the relevant counters or calculates the KPIs. Using the Threshold management system will define the threshold defines the threshold and whenever the threshold is crossed the Performance management system will send a threshold crossing alarm using the standard TM Forum Alarm management API. Other consequences (not just alarms) may also be defined in a textual format.

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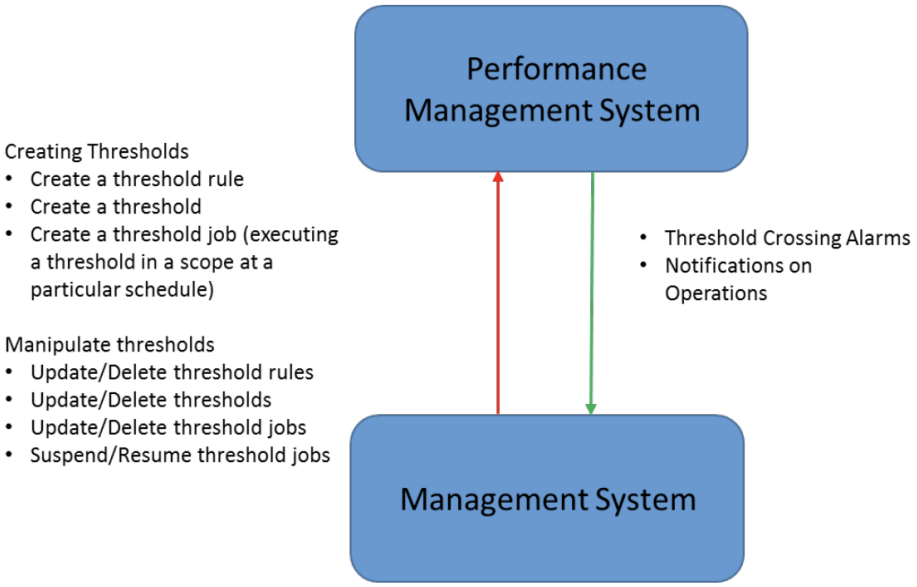


Figure 1 - Threshold Management Use Case

Threshold Definition Steps

A threshold is defined and executed in the target Performance management system in following steps:

1. Defining a threshold rule
2. Defining a threshold
3. Defining a threshold job that will execute the threshold with a scope of monitored objects (resources, services, and customers) at a specific schedule.

Threshold Rule Types

The Threshold Rule is the heart of this interface. It allows to define logic that determines when to alert on exceptional values of KPIs and on return to normal conditions. The Thresholding API supports two kinds of thresholds:

- Simple Thresholds – defined explicitly by a set of attributes
- Threshold algorithms – a case where the threshold is actually an algorithm to be executed with parameters

The simple threshold rules provide support for simple cases such as comparing a value of a KPI to a constant, but it can also support more interesting options such as checking ranges of values or counting events in a particular time interval and more.

It is also possible to define an action that will be triggered when a threshold rule is valid. In most cases the action will be raising or clearing a threshold crossing alarm (using the Alarm management interface). In other cases a generic kind of an action, described by a text or a URL can be defined.

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The Threshold resource packs a set of threshold rules to single logic (with an "OR" condition among them) such that hysteresis or multi-range thresholds can be supported.

Threshold Job

The Threshold Job is the operational entity responsible for executing a threshold, which consists of a set of threshold rules. It evaluates a collection of monitored objects—such as resources, services, and customers—at specific intervals as determined by the job's schedule. The job is designed to trigger alerts or actions when certain conditions or thresholds are met, ensuring that system performance, service levels, or other criteria are continuously monitored.

As part of defining a Threshold Job, two key elements are specified:

1. Scope of Monitored Objects:

```
This defines the set of resources, services, and customers that will be
monitored by the job. The monitored objects are subject to threshold evaluations
based on the rules set during job configuration.
```

2. Schedule:

```
A specific schedule is set for when the threshold job will run and evaluate the
defined thresholds. This schedule could be based on fixed intervals (e.g.,
hourly, daily) or specific times.
```

Execution States

Each Threshold Job has an associated Execution State, which represents the status or condition of the job at any given time during its lifecycle. The possible execution states for a threshold job are as follows:

- Active:

```
The job is currently active and running according to its schedule. This state
indicates that the job is ongoing and performing the threshold checks.
```

- Suspended:

```
The job has been suspended either manually or due to certain conditions that
temporarily stop its execution. The job is not executing during this state, but
it can be resumed later.
```

- Failed:

```
The job encountered an error during its execution, causing it to fail. The
failure might result from an issue with the job itself (e.g., misconfiguration,
```

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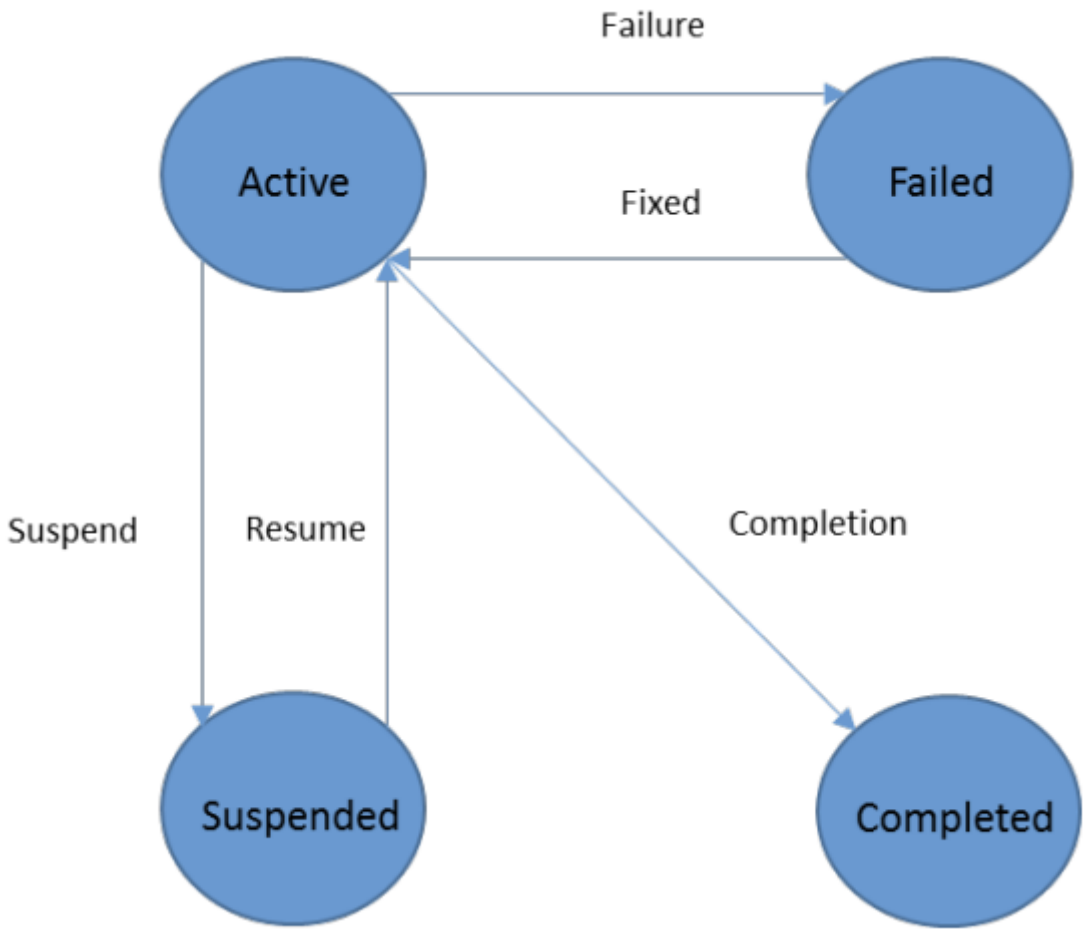
```
system failure) or from the monitored objects not meeting the required
thresholds. The job is not active during this state.
```

- Completed:

```
The job has successfully completed its execution. All the threshold checks have
been performed, and the job has finished its task. It is no longer active, but
it has successfully completed its intended work.
```

State Machine

A state machine diagram illustrates how the threshold job transitions through these execution states, depending on events such as job initiation, error occurrence, or completion of tasks. The diagram would typically show transitions between Active, Suspended, Failed, and Completed states in response to triggers like job start, error detection, or successful completion.



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Support of polymorphism and extension patterns

Overview

Support for polymorphic collections, types, and schema-based extensions is provided through a set of generic meta-attributes, described below. Polymorphism in collections occurs when entities inherit from base entities—for example, the rule types SimpleThresholdRule and AlgorithmThresholdRule inheriting properties from a common base entity that is ThresholdRule.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

Meta-Attributes

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Characteristic instances some may be instances of StringCharacteristic where other could be instances of IntegerCharacteristic. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an PerformanceMeasurementRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as PerformanceMeasurementRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

Threshold resource

A threshold base entity containing threshold rules.

Resource model

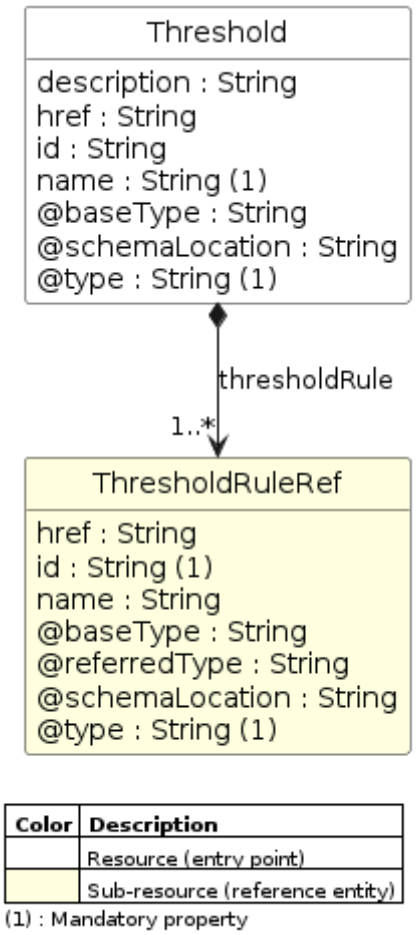


Figure 1 - Threshold

Field descriptions

Threshold fields

description	A String. A description of the threshold.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A word, term, or phrase by which a Performance threshold is known and distinguished from other thresholds.

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thresholdRule	A ThresholdRuleRef. A Performance Threshold contains a set of Performance threshold rules of different conditions (Raise, Clear) and different severities.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ThresholdRuleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'Threshold' resource object.


```
{
  "@type": "Threshold",
  "@baseType": "Threshold",
  "@schemaLocation": "/schema-registry/v5/Threshold.yml",
  "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-th-01",
  "id": "f3a34f89-0046-th-01",
  "description": "Network performance threshold for monitoring packet drop rate and
triggering alarms when acceptable limits are exceeded",
  "name": "threshold-1",
  "thresholdRule": [
    {
      "@type": "ThresholdRuleRef",
      "@schemaLocation": "/schema-registry/v5/ThresholdRuleRef.yml",
      "@referredType": "SimpleThresholdRule",
      "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-
01",
      "id": "f3a34f89-0046-thr-01"
    }
  ]
}
```


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ThresholdRule resource

ThresholdRule A base entity for concrete definition of thresholds. This datatype is abstract.

Resource model

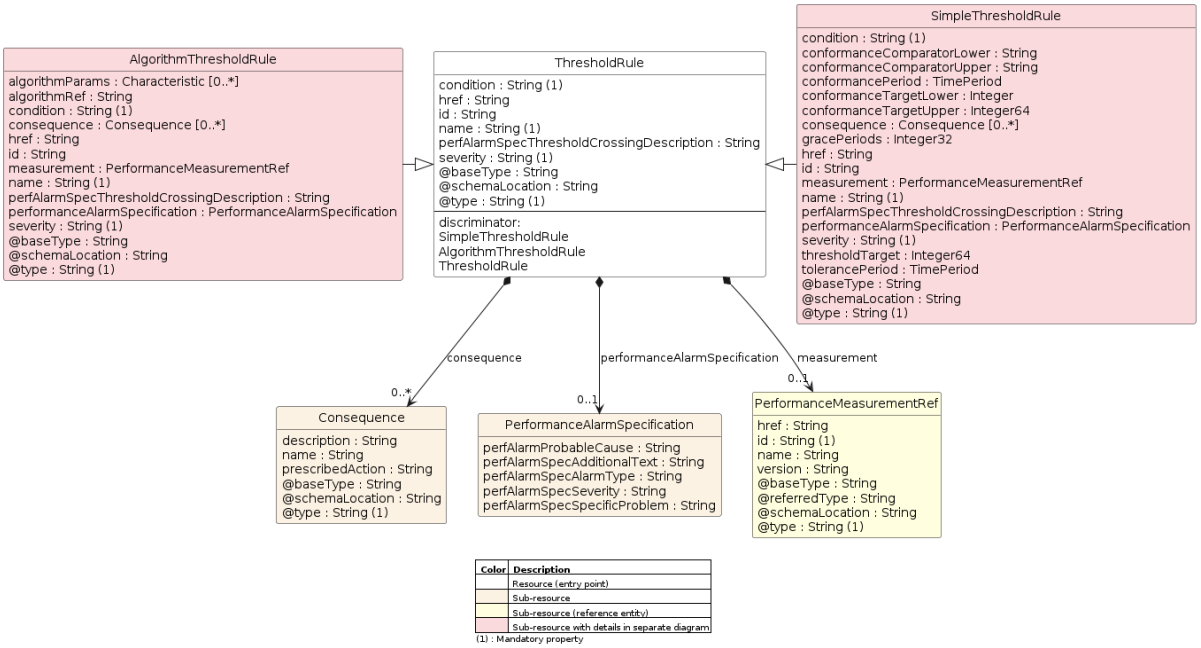


Figure 2 - ThresholdRule

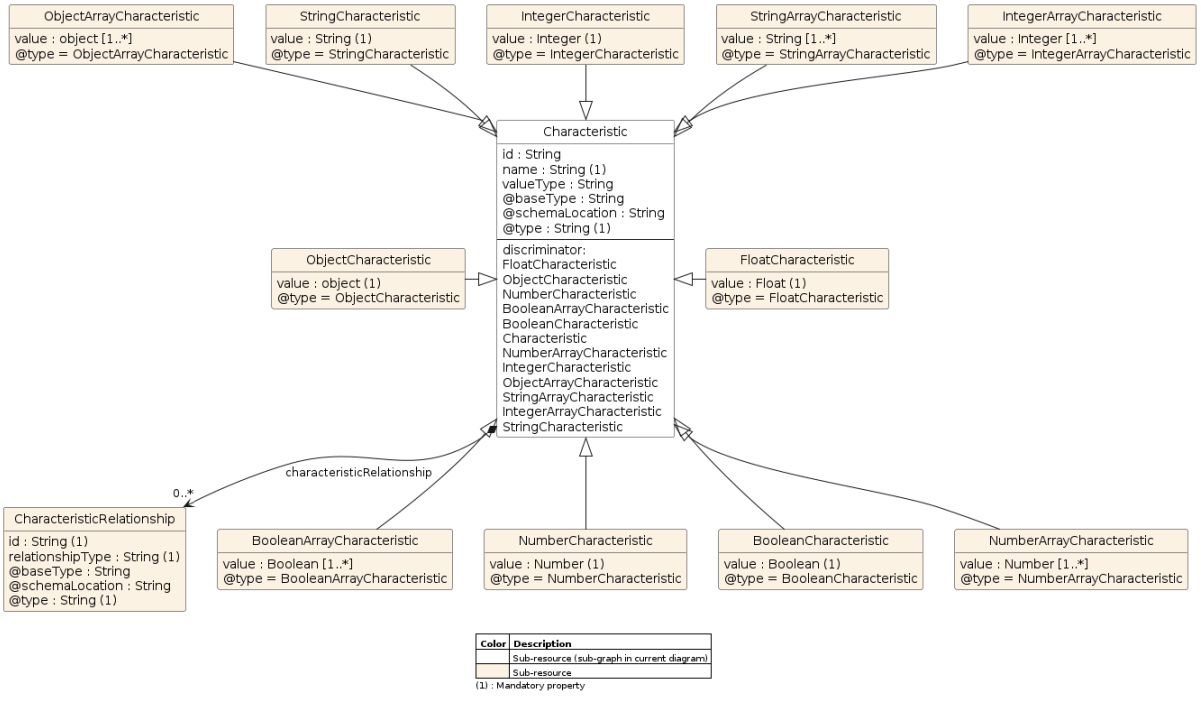


Figure 3 - Characteristic

Field descriptions

ThresholdRule fields

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condition	A String. A concrete threshold may have two possible values: 'Raise' - a threshold was crossed or 'Clear' - a threshold ceased crossing.
consequence	A Consequence. A threshold crossing or a threshold ceased to be crossing results in a Performance consequence.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
measurement	A PerformanceMeasurementRef. PerformanceMeasurement reference: A PerformanceMeasurement is the observed value(s) of some PerformanceIndicatorSpecification.
name	A String. A word, term, or phrase by which a Performance threshold rule is known and distinguished from other threshold rules.
perfAlarmSpecThresholdCrossing Description	A String. Defines the specification for threshold crossing alarms, including severity, type, probable cause, and additional context used when generating alarms.
performanceAlarmSpecification	A PerformanceAlarmSpecification. The specification of how to populate the alarm fields when generating a threshold crossing alarm.
severity	A String. A threshold can be generated in different severity levels. A crossing for each level may require a different condition and possibly trigger a different consequence. The supported severity levels are equivalent to the possible severity level of alarms. Examples are CRITICAL,MAJOR,MINOR,WARNING,INTERMEDIATE.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ThresholdRule can be instantiated as * AlgorithmThresholdRule * SimpleThresholdRule
algorithmParams	This property is present in subclasses
algorithmRef	This property is present in subclasses
conformanceComparatorLower	This property is present in subclasses
conformanceComparatorUpper	This property is present in subclasses
conformancePeriod	This property is present in subclasses
conformanceTargetLower	This property is present in subclasses
conformanceTargetUpper	This property is present in subclasses
gracePeriods	This property is present in subclasses
thresholdTarget	This property is present in subclasses

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tolerancePeriod	This property is present in subclasses

Consequence sub-resource fields

description	A String. A description of a Consequence.
name	A String. A word, term, or phrase by which a Consequence is known and distinguished from other measurements.
prescribedAction	A String. The suggested action.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceMeasurementRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the PerformanceMeasurement.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceAlarmSpecification sub-resource fields

perfAlarmProbableCause	A String. A value for the ProbableCause attribute as part of a threshold crossing alarm definition.
perfAlarmSpecAdditionalText	A String. A value for the AdditionalText attribute as part of a threshold crossing alarm definition.
perfAlarmSpecAlarmType	A String. A value for the AlarmType attribute as part of a threshold crossing alarm definition.
perfAlarmSpecSeverity	A String. A value for the Severity attribute as part of a threshold crossing alarm definition. Examples are CRITICAL,MAJOR,MINOR,WARNING,INTERMEDIATE.
perfAlarmSpecSpecificProblem	A String. A value for the SpecificProblem attribute as part of a threshold crossing alarm definition.

Characteristic sub-resource fields

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characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. A characteristic which value is an array of Boolean(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. A characteristic which value is an array of Integer(s).
valueType	A String. Data type of the value of the characteristic.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. A characteristic which value is an array of Number(s).
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.

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valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Collection of characteristic values.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.

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value	A String. Collection of string characteristics.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ThresholdRule' resource object.


```
{
  "@type": "SimpleThresholdRule",
  "@baseType": "ThresholdRule",
  "@schemaLocation": "/schema-registry/v5/SimpleThresholdRule.yml",
  "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-01",
  "id": "f3a34f89-0046-thr-01",
  "measurement": {
    "@type": "PerformanceMeasurementRef",
    "@schemaLocation": "/schema-registry/v5/PerformanceMeasurementRef.yaml",
    "href": "tmf-api/performanceManagement/v5/performanceMeasurement/f3a34f89-0046-
msr-01",
    "id": "f3a34f89-0046-msr-01",
    "@referredType": "PerformanceMeasurement"
  },
  "consequence": [
    {
      "@type": "Consequence",
      "@baseType": "Consequence",
      "@schemaLocation": "/schema-registry/v5/Consequence.yaml",
      "description": "Raises a major severity alarm when packet drop rate exceeds
the defined threshold of 283 packets per second",
      "name": "Alarm Consequence",
```


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```
      "prescribedAction": "CreateAlarm"
    }
  ],
  "perfAlarmSpecThresholdCrossingDescription": "Packet drop rate threshold violation
detected - dropped packets per second exceeded acceptable limit",
  "performanceAlarmSpecification": {
    "perfAlarmSpecAdditionalText": "The packet drop rate has exceeded the defined
performance threshold, indicating potential congestion or network instability. This
condition may lead to degraded application performance or increased latency if not
resolved promptly.",
    "perfAlarmSpecAlarmType": "PERFORMANCE_DEGRADATION",
    "perfAlarmSpecSeverity": "MAJOR",
    "perfAlarmSpecSpecificProblem": "Packet drop rate exceeded",
    "perfAlarmProbableCause": "Threshold Crossed"
  },
  "condition": "Raise",
  "name": "DropPacketsHighRule",
  "severity": "MAJOR",
  "conformanceComparatorLower": "GT",
  "conformanceComparatorUpper": "LT",
  "conformancePeriod": {
    "startDateTime": "2025-04-10T06:38:38Z",
    "endDateTime": "2025-04-17T06:38:38Z"
  },
  "conformanceTargetLower": 132,
  "conformanceTargetUpper": 776,
  "gracePeriods": 933,
  "thresholdTarget": 283,
  "tolerancePeriod": {
    "startDateTime": "2025-04-01T06:38:38Z",
    "endDateTime": "2025-04-10T06:38:38Z"
  }
}
```


We provide below a JSON representation as example of the 'ThresholdRule' resource object.


```
{
  "@type": "AlgorithmThresholdRule",
  "@baseType": "ThresholdRule",
  "@schemaLocation": "/schema-registry/v5/AlgorithmThresholdRule.yaml",
  "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-01",
  "id": "f3a34f89-0046-thr-01",
  "measurement": {
    "@type": "PerformanceMeasurementRef",
    "@baseType": "PerformanceMeasurementRef",
    "@schemaLocation": "/schema-registry/v5/PerformanceMeasurementRef.yaml",
    "href": "tmf-api/performanceManagement/v5/performanceMeasurement/f3a34f89-0046-
msr-01",
    "id": "f3a34f89-0046-msr-01",
    "@referredType": "PerformanceMeasurement"
  },
  "consequence": [
    {
      "@type": "Consequence",
      "@baseType": "Consequence",
      "@schemaLocation": "/schema-registry/v5/Consequence.yaml",
      "description": "Raises a major severity alarm when algorithmic analysis
detects packet drop rate anomaly exceeding adaptive threshold",
      "name": "Alarm Consequence",
```


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```
      "prescribedAction": "CreateAlarm"
    }
  ],
  "perfAlarmSpecThresholdCrossingDescription": "Algorithm-based packet drop rate
threshold violation detected - anomalous pattern identified",
  "performanceAlarmSpecification": {
    "perfAlarmSpecAdditionalText": "The algorithmic threshold rule has detected an
abnormal packet drop rate pattern based on dynamic analysis with sensitivity factor
0.2. This indicates potential network congestion, equipment malfunction, or traffic
anomalies requiring immediate investigation.",
    "perfAlarmSpecAlarmType": "PERFORMANCE_DEGRADATION",
    "perfAlarmSpecSeverity": "MAJOR",
    "perfAlarmSpecSpecificProblem": "Packet drop rate exceeded",
    "perfAlarmProbableCause": "Threshold Crossed"
  },
  "condition": "Raise",
  "name": "DropPacketsHighRule",
  "severity": "MAJOR",
  "algorithmParams": [
    {
      "id": "1",
      "name": "sensitivity",
      "valueType": "float",
      "value": 0.2,
      "@type": "FloatCharacteristic",
      "@baseType": "Characteristic",
      "@schemaLocation": "/schema-registry/v5/FloatCharacteristic.schema.yaml"
    }
  ],
  "algorithmRef": "http://api/thresholdrulealgoritm/7000"
}
```


ThresholdJob resource

The Threshold Job is the operational entity to execute a threshold (a set of threshold rules). As part of the definition of a Threshold Job, a scope of monitored objects is provided and a schedule for the threshold evaluation is determined.

Resource model

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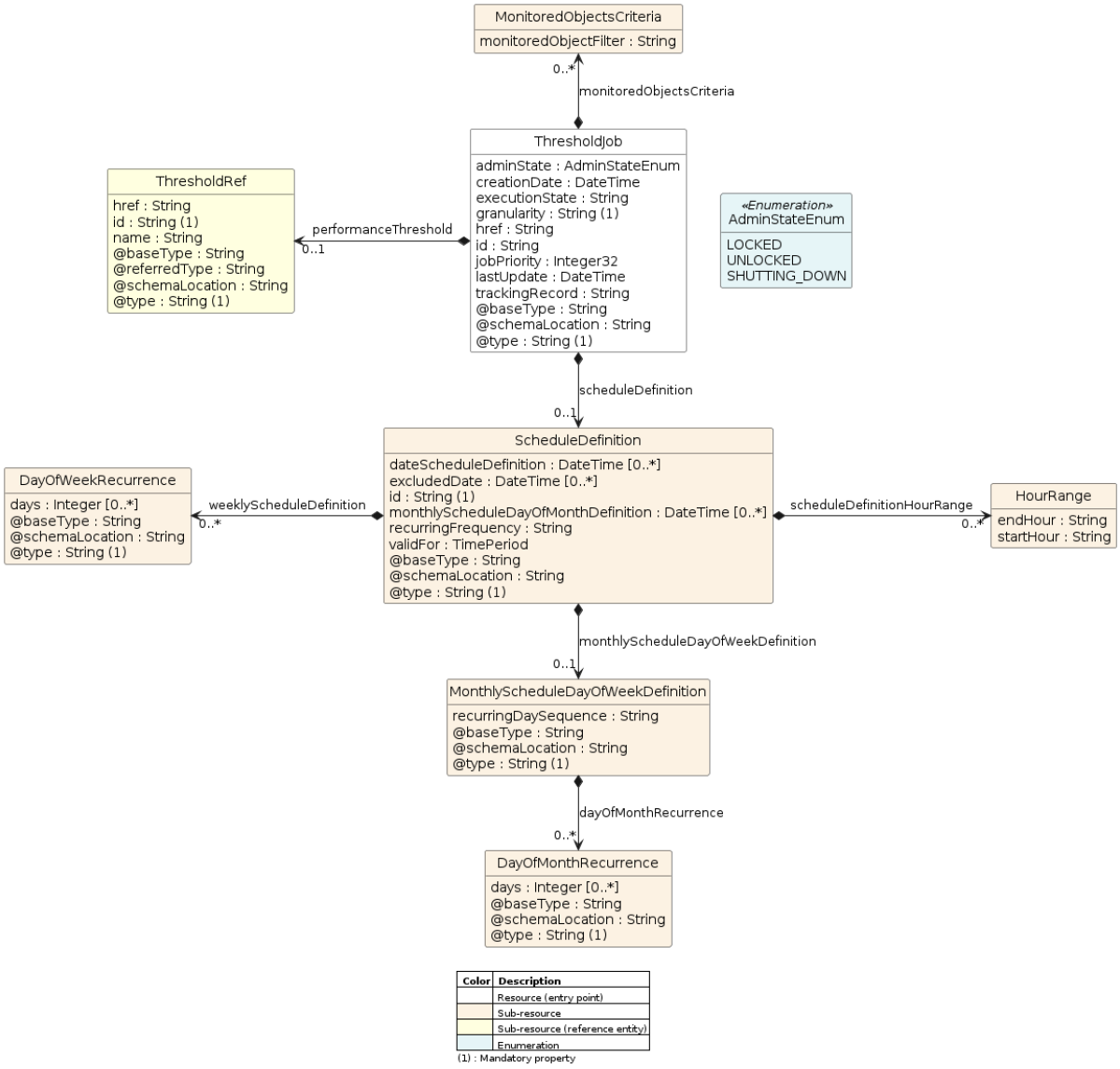


Figure 4 - ThresholdJob

Field descriptions

ThresholdJob fields

adminState	An AdminStateEnum. Enumeration of administrative states of a resource. Possible values indicate whether the resource is locked, unlocked, or shutting down. ENUMERATED with values: * LOCKED * UNLOCKED * SHUTTING_DOWN
creationDate	A DateTime. Timestamp indicating when the Threshold Job was originally created in the Performance Management system.

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executionState	A String. The current runtime state of the Threshold Job. Supported states: 'Active' - The job is currently active and executing threshold evaluation. 'Suspended' - The job is temporarily stopped but can be resumed. 'Failed' - The job stopped due to a failure. 'Completed' - The job has finished its execution successfully and is no longer active.
granularity	A String. Defines the evaluation granularity of the Threshold Job (e.g., G_5MN, G_15MN, G_30MN, G_1H). It specifies the periodic time interval at which KPIs are assessed against threshold rules.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
jobPriority	An Integer32. Relative priority level assigned to the Threshold Job. Higher priority jobs may be executed or scheduled ahead of lower priority jobs.
lastUpdate	A DateTime. Timestamp of the most recent update or modification to the Threshold Job definition.
monitoredObjectsCriteria	A MonitoredObjectsCriteria. Criteria that define the scope of monitored objects (resources, services, or customers) to which the Threshold Job applies.
performanceThreshold	A ThresholdRef.
scheduleDefinition	A ScheduleDefinition. The schedule definition for running jobs.
trackingRecord	A String. Optional field to store tracking or correlation information related to the Threshold Job (e.g., audit trails, external references, or operator notes).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MonitoredObjectsCriteria sub-resource fields

monitoredObjectFilter	A String. A filter that can be used in conjunction with the monitored object class for specifying the set of instances that are referenced.

ThresholdRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ScheduleDefinition sub-resource fields

dateScheduleDefinition	A DateTime. The date schedule is used to define a schedule that is based on specific dates and times. Each value must be a valid IETF RFC 3339 date-time string (e.g., '2015-12-31T23:59:59Z', '2013-02-28T09:00:00-05:00').
excludedDate	A DateTime. A list of specific dates that should be excluded from the Schedule Definition.
id	A String. Identifier of the schedule definition, it should be locally unique.
monthlyScheduleDayOfMonthDefinition	A DateTime. The schedule definition for running the threshold job.
monthlyScheduleDayOfWeekDefinition	A MonthlyScheduleDayOfWeekDefinition.
recurringFrequency	A String. A recurring frequency to run a job within day that is included in schedule definition, for example: every 5 minutes, 15 minute, 30 minutes, 1 hour.
scheduleDefinitionHourRange	A HourRange. A list of time ranges within a specific day that the schedule will be active on, for example 08:00-12:00, 16:00-19:00.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
weeklyScheduleDefinition	A DayOfWeekRecurrence. The weekly schedule is used to define a schedule that is based on the days of the week, e.g. a schedule that will be active only on Monday and Tuesday.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MonthlyScheduleDayOfWeekDefinition sub-resource fields

dayOfMonthRecurrence	A DayOfMonthRecurrence. Specifies recurrence on particular days of the month (1?31).
recurringDaySequence	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

DayOfMonthRecurrence sub-resource fields

days	An Integer. List of days in the month on which recurrence occurs.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

HourRange sub-resource fields

endHour	A String. The end of the hour range, expressed in 24-hour HH:mm format (e.g., 12:00 for noon, 23:59 for one minute before midnight).
startHour	A String. The beginning of the hour range, expressed in 24-hour HH:mm format (e.g., 09:00 for 9 AM, 17:30 for 5:30 PM).

DayOfWeekRecurrence sub-resource fields

days	An Integer. List of days in the week on which recurrence occurs.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ThresholdJob' resource object.


```
{
  "@type": "ThresholdJob",
  "adminState": "LOCKED",
  "creationDate": "2025-02-12T06:37:22Z",
  "executionState": "Active",
  "granularity": "G_30MN",
  "id": "3a34f89-0046-tjb-01",
  "href": "tmf-api/performanceThresholding/v5/thresholdJob/3a34f89-0046-tjb-01",
  "jobPriority": 610,
  "lastUpdate": "2025-02-12T06:37:22Z",
  "monitoredObjectsCriteria": [
    {
      "monitoredObjectFilter": "nodeName=Router01"
    }
  ],
  "performanceThreshold": {
    "@type": "ThresholdRef",
```


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```
    "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-pfth-01",
    "id": "f3a34f89-0046-pfth-01",
    "@referredType": "Threshold"
  },
  "scheduleDefinition": {
    "@type": "ScheduleDefinition",
    "@baseType": "ScheduleDefinition",
    "@schemaLocation": "/schema-registry/v5/ScheduleDefinition.yaml",
    "id": "f3a34f89-0046-sd-01",
    "validFor": {
      "startDateTime": "2025-04-01T06:37:22Z",
      "endDateTime": "2025-04-15T06:37:22Z"
    },
    "recurringFrequency": "every 5 minutes",
    "excludedDate": [
      "2025-02-12T06:37:22Z"
    ],
    "scheduleDefinitionHourRange": [
      {
        "startHour": "09:00",
        "endHour": "17:30"
      }
    ],
    "weeklyScheduleDefinition": [],
    "monthlyScheduleDayOfMonthDefinition": [
      "2025-04-01T06:37:22Z",
      "2025-02-15T06:37:22Z"
    ],
    "monthlyScheduleDayOfWeekDefinition": {
      "@type": "MonthlyScheduleDayOfWeekDefinition",
      "@baseType": "MonthlyScheduleDayOfWeekDefinition",
      "@schemaLocation": "/schema-
registry/v5/MonthlyScheduleDayOfWeekDefinition.yaml",
      "dayOfMonthRecurrence": [
        {
          "@type": "DayOfMonthRecurrence",
          "@baseType": "DayOfMonthRecurrence",
          "@schemaLocation": "/schema-registry/v5/DayOfMonthRecurrence.yaml"
        }
      ],
      "recurringDaySequence": "FirstWeekOfMonth"
    }
  },
  "trackingRecord": "Router01_PacketDropRate_ThresholdMonitoring_ExecutionLog"
}
```


AlgorithmThresholdRule resource

A threshold rule that is a reference to an algorithm.

Resource model

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Figure 5 - AlgorithmThresholdRule

See Figure 3 for Characteristic diagrams.

Field descriptions

AlgorithmThresholdRule fields

algorithmParams	A Characteristic. Collection of algorithm Params. Characteristic can be instantiated as * BooleanArrayCharacteristic * BooleanCharacteristic * FloatCharacteristic * IntegerArrayCharacteristic * IntegerCharacteristic * NumberArrayCharacteristic * NumberCharacteristic * ObjectArrayCharacteristic * ObjectCharacteristic * StringArrayCharacteristic * StringCharacteristic
algorithmRef	A String. The href of the algorithm for threshold rule.
condition	A String. A concrete threshold may have two possible values: 'Raise' - a threshold was crossed or 'Clear' - a threshold ceased crossing.
consequence	A Consequence. A threshold crossing or a threshold ceased to be crossing results in a Performance consequence.

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href	A String. Hyperlink reference.
id	A String. Unique identifier.
measurement	A PerformanceMeasurementRef. PerformanceMeasurement reference: A PerformanceMeasurement is the observed value(s) of some PerformanceIndicatorSpecification.
name	A String. A word, term, or phrase by which a Performance threshold rule is known and distinguished from other threshold rules.
perfAlarmSpecThresholdCrossing Description	A String. Defines the specification for threshold crossing alarms, including severity, type, probable cause, and additional context used when generating alarms.
performanceAlarmSpecification	A PerformanceAlarmSpecification. The specification of how to populate the alarm fields when generating a threshold crossing alarm.
severity	A String. A threshold can be generated in different severity levels. A crossing for each level may require a different condition and possibly trigger a different consequence. The supported severity levels are equivalent to the possible severity level of alarms. Examples are CRITICAL,MAJOR,MINOR,WARNING,INTERMEDIATE.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

See [Characteristic] for details.

CharacteristicRelationship sub-resource fields

See [CharacteristicRelationship] for details.

BooleanArrayCharacteristic sub-resource fields

See [BooleanArrayCharacteristic] for details.

BooleanCharacteristic sub-resource fields

See [BooleanCharacteristic] for details.

FloatCharacteristic sub-resource fields

See [FloatCharacteristic] for details.

IntegerArrayCharacteristic sub-resource fields

See [IntegerArrayCharacteristic] for details.

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IntegerCharacteristic sub-resource fields

See [IntegerCharacteristic] for details.

NumberArrayCharacteristic sub-resource fields

See [NumberArrayCharacteristic] for details.

NumberCharacteristic sub-resource fields

See [NumberCharacteristic] for details.

ObjectArrayCharacteristic sub-resource fields

See [ObjectArrayCharacteristic] for details.

ObjectCharacteristic sub-resource fields

See [ObjectCharacteristic] for details.

StringArrayCharacteristic sub-resource fields

See [StringArrayCharacteristic] for details.

StringCharacteristic sub-resource fields

See [StringCharacteristic] for details.

Consequence sub-resource fields

See [Consequence] for details.

PerformanceMeasurementRef sub-resource fields

See [PerformanceMeasurementRef] for details.

PerformanceAlarmSpecification sub-resource fields

See [PerformanceAlarmSpecification] for details.

SimpleThresholdRule resource

A threshold rule that is based on explicit definition of attributes that represent a Boolean JsonLdExpression.

Resource model

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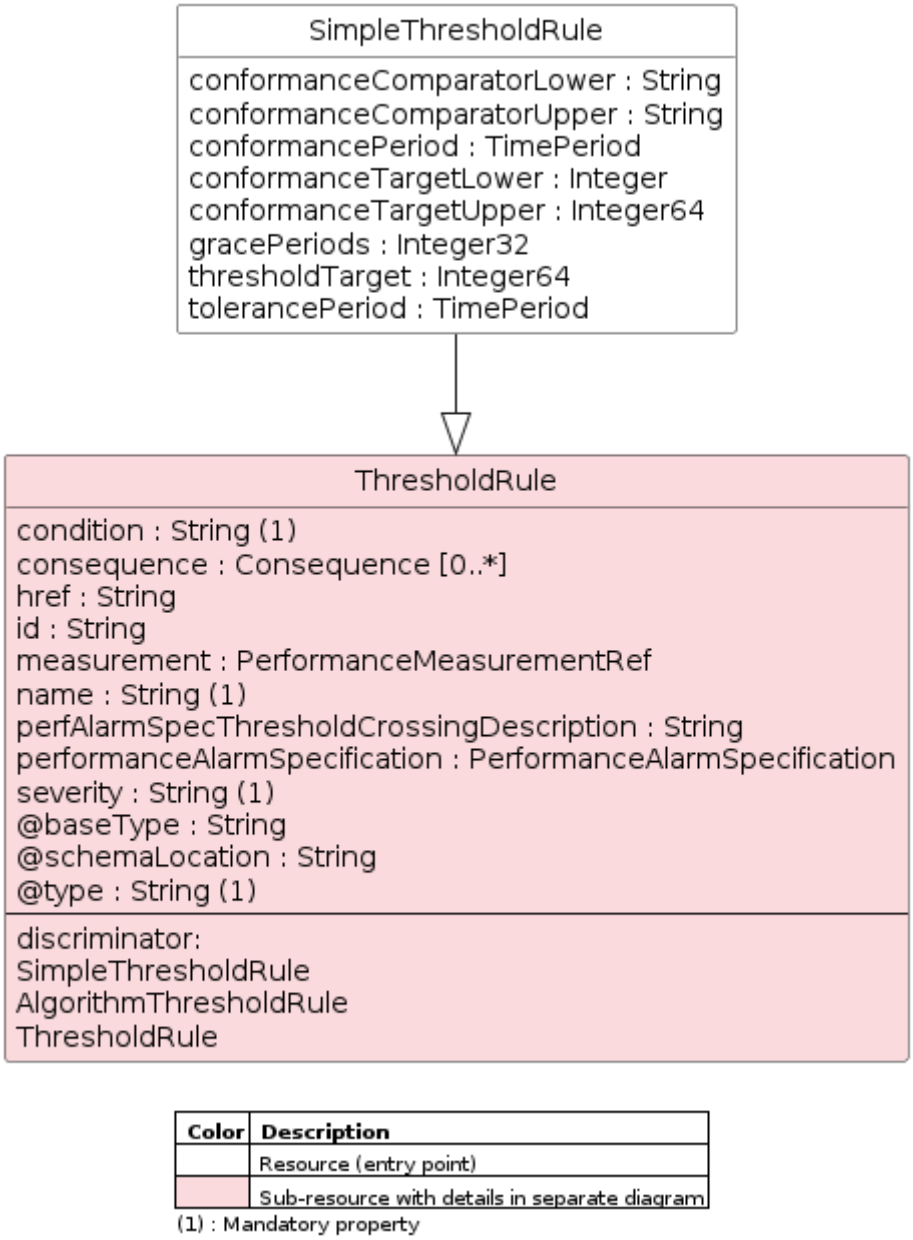


Figure 6 - SimpleThresholdRule

Field descriptions

SimpleThresholdRule fields

condition	A String. A concrete threshold may have two possible values: 'Raise' - a threshold was crossed or 'Clear' - a threshold ceased crossing.
conformanceComparatorLower	A String. An operator that when applied on a value (of conformanceTargetLower) specifies whether a Performance threshold rule is crossed or ceased to be crossed. Supported Conformance Operators are as follows: 'GT' for Greater than, 'GE' for Greater or equal, 'EQ' for Equal, 'NEQ' for Not Equal, and 'LE' for Less or equal.

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conformanceComparatorUpper	A String. An operator that when applied on a value (of conformanceTargetUpper) specifies whether a Performance threshold rule is crossed or ceased to be crossed. Supported Conformance Operators are as follows: 'GT' for Greater than, 'GE' for Greater or equal, 'EQ' for Equal, 'NEQ' for Not Equal, and 'LE' for Less or equal.
conformancePeriod	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
conformanceTargetLower	An Integer. A value used to determine if the Performance threshold is crossed or ceased to be crossed, serving as an lower value.
conformanceTargetUpper	An Integer64. A value used to determine if the Performance threshold is crossed or ceased to be crossed, serving as an upper value.
consequence	A Consequence. A threshold crossing or a threshold ceased to be crossing results in a Performance consequence.
gracePeriods	An Integer32. The number of times an objective can remain un-updated in reference to the conformancePeriod without a Performance threshold Consequence being initiated.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
measurement	A PerformanceMeasurementRef. PerformanceMeasurement reference: A PerformanceMeasurement is the observed value(s) of some PerformanceIndicatorSpecification.
name	A String. A word, term, or phrase by which a Performance threshold rule is known and distinguished from other threshold rules.
perfAlarmSpecThresholdCrossing Description	A String. Defines the specification for threshold crossing alarms, including severity, type, probable cause, and additional context used when generating alarms.
performanceAlarmSpecification	A PerformanceAlarmSpecification. The specification of how to populate the alarm fields when generating a threshold crossing alarm.
severity	A String. A threshold can be generated in different severity levels. A crossing for each level may require a different condition and possibly trigger a different consequence. The supported severity levels are equivalent to the possible severity level of alarms. Examples are CRITICAL,MAJOR,MINOR,WARNING,INTERMEDIATE.
thresholdTarget	An Integer64. A percent that is used to specify when a warning should be used that indicates a Performance threshold is in danger of not being met.
tolerancePeriod	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Consequence sub-resource fields

See [Consequence] for details.

PerformanceMeasurementRef sub-resource fields

See [PerformanceMeasurementRef] for details.

PerformanceAlarmSpecification sub-resource fields

See [PerformanceAlarmSpecification] for details.

Notification Resource Models

10 notifications are defined for this API.

Notifications related to Threshold:

- Create Event
- Delete Event
- Attribute Value Change Event

Notifications related to ThresholdRule:

- Create Event
- Delete Event
- Attribute Value Change Event

Notifications related to ThresholdJob:

- Create Event
- Delete Event
- Attribute Value Change Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

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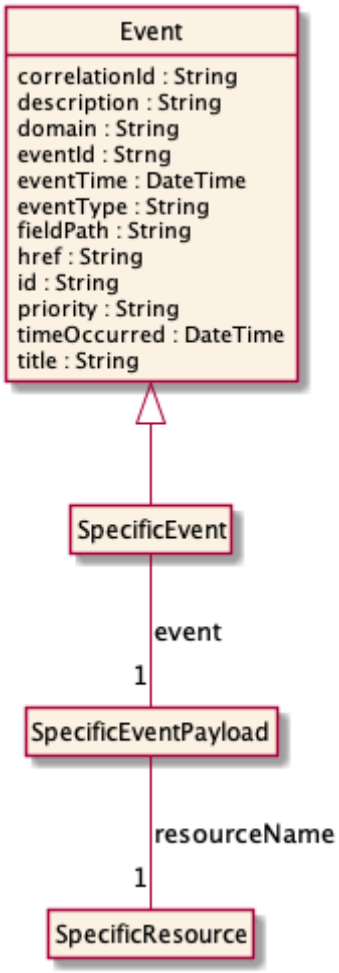


Figure 7 Notification Pattern

Threshold

Create Event

Message example for ThresholdCreateEvent event

```
Content-Type: application/json
{
  "@type": "ThresholdCreateEvent",
  "@baseType": "ThresholdCreateEvent",
  "@schemaLocation": "/schema-registry/ThresholdCreateEvent.yaml",
  "href": "/tmf-api/performanceManagement/ThresholdCreateEvent/f3a34f89-0046-tjb-
01",
  "id": "f3a34f89-0046-01",
  "correlationId": "f3a34f89-0046-cr-01",
  "domain": "Commercial",
  "title": "ThresholdCreateEvent",
  "description": "ThresholdCreateEvent illustration",
  "priority": "5",
  "timeOccurred": "2024-11-18T09:35:14Z",
  "reportingSystem": {
    "@type": "SoftwareRef",
    "@schemaLocation":
"https://mycsp.com:8080/tmfapi/schema/SoftwareRef/Extended.schema.json",
```

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```
    "@referredType": "SoftwareResource",
    "id": "533242BC35DSSA55555",
    "href":
"https://mycsp.com:8080/tmfapi/resourceManagement/v5/resource/533242BC35DSSA55555",
    "name": "BlueSkySoftware BSS/OSS"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "@schemaLocation": "/schema-registry/PartyRoleRef.yaml",
        "href": "/tmf-api/customerManagement/v5/customer/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "John Doe",
        "@referredType": "Customer",
        "partyId": "f3a34f89-0046-pt-01",
        "partyName": "John Doe"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/PartyRef.yaml",
        "href": "/tmf-api/partyManagement/v5/organization/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "6d Tech",
        "@referredType": "Organization"
      }
    }
  ],
  "eventId": "f3a34f89-0046-evtid-01",
  "eventTime": "2024-11-18T09:35:14Z",
  "eventType": "ThresholdCreateEvent",
  "event": {
    "threshold": {
      "@type": "Threshold",
      "@baseType": "Threshold",
      "@schemaLocation": "/schema-registry/v5/Threshold.yml",
      "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-th-01",
      "id": "f3a34f89-0046-th-01",
      "description": "Network performance threshold for monitoring packet drop rate
and triggering alarms when acceptable limits are exceeded",
      "name": "threshold-1",
      "thresholdRule": [
        {
          "@type": "ThresholdRuleRef",
          "@schemaLocation": "/schema-registry/v5/ThresholdRuleRef.yml",
          "@referredType": "SimpleThresholdRule",
          "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-
thr-01",
          "id": "f3a34f89-0046-thr-01"
        }
      ]
    }
  }
}
```


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Delete Event

Message example for ThresholdDeleteEvent event

```
Content-Type: application/json
{
  "@type": "ThresholdDeleteEvent",
  "@baseType": "ThresholdDeleteEvent",
  "@schemaLocation": "/schema-registry/ThresholdDeleteEvent.yaml",
  "href": "/tmf-api/performanceManagement/ThresholdDeleteEvent/f3a34f89-0046-tjb-
01",
  "id": "f3a34f89-0046-01",
  "correlationId": "f3a34f89-0046-cr-01",
  "domain": "Commercial",
  "title": "ThresholdDeleteEvent",
  "description": "ThresholdDeleteEvent illustration",
  "priority": "5",
  "timeOccurred": "2024-11-18T09:35:14Z",
  "source": {
    "@type": "EntityRef",
    "@schemaLocation": "/schema-registry/EntityRef.yaml",
    "href": "/tmf-api/resourceInventoryManagement/v5/logicalResource/f3a34f89-0046-
src-01",
    "id": "f3a34f89-0046-src-01",
    "name": "CRM App",
    "@referredType": "LogicalResource"
  },
  "reportingSystem": {
    "@type": "SoftwareRef",
    "@schemaLocation":
"https://mycsp.com:8080/tmfapi/schema/SoftwareRef/Extended.schema.json",
    "@referredType": "SoftwareResource",
    "id": "533242BC35DSSA55555",
    "href":
"https://mycsp.com:8080/tmfapi/resourceManagement/v5/resource/533242BC35DSSA55555",
    "name": "BlueSkySoftware BSS/OSS"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "@schemaLocation": "/schema-registry/PartyRoleRef.yaml",
        "href": "/tmf-api/customerManagement/v5/customer/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "John Doe",
        "@referredType": "Customer",
        "partyId": "f3a34f89-0046-pt-01",
        "partyName": "John Doe"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/PartyRef.yaml",
        "href": "/tmf-api/partyManagement/v5/organization/f3a34f89-0046-pt-01",
```

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```
        "id": "f3a34f89-0046-pt-01",
        "name": "6d Tech",
        "@referredType": "Organization"
      }
    }
  ],
  "eventId": "f3a34f89-0046-evtid-01",
  "eventTime": "2024-11-18T09:35:14Z",
  "eventType": "ThresholdDeleteEvent",
  "event": {
    "threshold": {
      "@type": "Threshold",
      "@baseType": "Threshold",
      "@schemaLocation": "/schema-registry/v5/Threshold.yml",
      "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-th-01",
      "id": "f3a34f89-0046-th-01"
    }
  }
}
```


Attribute Value Change Event

Message example for ThresholdAttributeValueChangeEvent event


```
Content-Type: application/json
{
  "@type": "ThresholdAttributeValueChangeEvent",
  "@baseType": "ThresholdAttributeValueChangeEvent",
  "@schemaLocation": "/schema-registry/ThresholdAttributeValueChangeEvent.yaml",
  "href": "/tmf-
api/performanceManagement/ThresholdAttributeValueChangeChange/f3a34f89-0046-tjb-01",
  "id": "f3a34f89-0046-01",
  "correlationId": "f3a34f89-0046-cr-01",
  "domain": "Commercial",
  "title": "ThresholdAttributeValueChangeEvent",
  "description": "ThresholdAttributeValueChangeEvent illustration",
  "priority": "5",
  "timeOccurred": "2024-11-18T09:35:14Z",
  "reportingSystem": {
    "@type": "SoftwareRef",
    "@schemaLocation":
"https://mycsp.com:8080/tmfapi/schema/SoftwareRef/Extended.schema.json",
    "@referredType": "SoftwareResource",
    "id": "533242BC35DSSA55555",
    "href":
"https://mycsp.com:8080/tmfapi/resourceManagement/v5/resource/533242BC35DSSA55555",
    "name": "BlueSkySoftware BSS/OSS"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "@schemaLocation": "/schema-registry/PartyRoleRef.yaml",
        "href": "/tmf-api/customerManagement/v5/customer/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "John Doe",
        "@referredType": "Customer",
```


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```
        "partyId": "f3a34f89-0046-pt-01",
        "partyName": "John Doe"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/PartyRef.yaml",
        "href": "/tmf-api/partyManagement/v5/organization/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "6d Tech",
        "@referredType": "Organization"
      }
    }
  ],
  "eventId": "f3a34f89-0046-evtid-01",
  "eventTime": "2024-11-18T09:35:14Z",
  "eventType": "ThresholdAttributeValueChangeEvent",
  "event": {
    "threshold": {
      "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-th-01",
      "id": "f3a34f89-0046-th-01",
      "@type": "Threshold",
      "description": "Network performance threshold for monitoring packet drop rate
and triggering alarms when acceptable limits are exceeded"
    }
  }
}
```


ThresholdRule

Create Event

Message example for ThresholdRuleCreateEvent event


```
Content-Type: application/json
{
  "@type": "ThresholdRuleCreateEvent",
  "@baseType": "ThresholdRuleCreateEvent",
  "@schemaLocation": "/schema-registry/ThresholdRuleChange.yaml",
  "href": "/tmf-api/performanceManagement/ThresholdRuleCreate/f3a34f89-0046-tjb-01",
  "id": "f3a34f89-0046-01",
  "correlationId": "f3a34f89-0046-cr-01",
  "domain": "Commercial",
  "title": "ThresholdRuleChangeEvent",
  "description": "ThresholdRuleChangeEvent illustration",
  "priority": "5",
  "timeOccurred": "2024-11-18T09:35:14Z",
  "source": {
    "@type": "EntityRef",
    "@schemaLocation": "/schema-registry/EntityRef.yaml",
    "href": "/tmf-api/resourceInventoryManagement/v5/logicalResource/f3a34f89-0046-
src-01",
    "id": "f3a34f89-0046-src-01",
    "name": "CRM App",
    "@referredType": "LogicalResource"
```


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```
  },
  "reportingSystem": {
    "@type": "SoftwareRef",
    "@schemaLocation":
"https://mycsp.com:8080/tmfapi/schema/SoftwareRef/Extended.schema.json",
    "@referredType": "SoftwareResource",
    "id": "533242BC35DSSA55555",
    "href":
"https://mycsp.com:8080/tmfapi/resourceManagement/v5/resource/533242BC35DSSA55555",
    "name": "BlueSkySoftware BSS/OSS"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "@schemaLocation": "/schema-registry/PartyRoleRef.yaml",
        "href": "/tmf-api/customerManagement/v5/customer/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "John Doe",
        "@referredType": "Customer",
        "partyId": "f3a34f89-0046-pt-01",
        "partyName": "John Doe"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/PartyRef.yaml",
        "href": "/tmf-api/partyManagement/v5/organization/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "6d Tech",
        "@referredType": "Organization"
      }
    }
  ],
  "eventId": "f3a34f89-0046-evtid-01",
  "eventTime": "2024-11-18T09:35:14Z",
  "eventType": "ThresholdRuleCreateEvent",
  "event": {
    "thresholdRule": {
      "@type": "SimpleThresholdRule",
      "@baseType": "ThresholdRule",
      "@schemaLocation": "/schema-registry/v5/SimpleThresholdRule.yml",
      "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-
01",
      "id": "f3a34f89-0046-thr-01",
      "measurement": {
        "@type": "PerformanceMeasurementRef",
        "@schemaLocation": "/schema-registry/v5/PerformanceMeasurementRef.yaml",
        "href": "tmf-api/performanceManagement/v5/performanceMeasurement/f3a34f89-
0046-msr-01",
        "id": "f3a34f89-0046-msr-01",
        "@referredType": "PerformanceMeasurement"
      },
      "consequence": [
        {
          "@type": "Consequence",
```


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```
          "@baseType": "Consequence",
          "@schemaLocation": "/schema-registry/v5/Consequence.yaml",
          "description": "Raises a major severity alarm when packet drop rate
exceeds the defined threshold of 283 packets per second",
          "name": "Alarm Consequence",
          "prescribedAction": "CreateAlarm"
        }
      ],
      "perfAlarmSpecThresholdCrossingDescription": "Packet drop rate threshold
violation detected - dropped packets per second exceeded acceptable limit",
      "performanceAlarmSpecification": {
        "perfAlarmSpecAdditionalText": "The packet drop rate has exceeded the
defined performance threshold, indicating potential congestion or network
instability. This condition may lead to degraded application performance or
increased latency if not resolved promptly.",
        "perfAlarmSpecAlarmType": "PERFORMANCE_DEGRADATION",
        "perfAlarmSpecSeverity": "MAJOR",
        "perfAlarmSpecSpecificProblem": "Packet drop rate exceeded",
        "perfAlarmProbableCause": "Threshold Crossed"
      },
      "condition": "Raise",
      "name": "DropPacketsHighRule",
      "severity": "MAJOR",
      "conformanceComparatorLower": "GT",
      "conformanceComparatorUpper": "LT",
      "conformancePeriod": {
        "startDateTime": "2025-04-10T06:38:38Z",
        "endDateTime": "2025-04-17T06:38:38Z"
      },
      "conformanceTargetLower": 132,
      "conformanceTargetUpper": 776,
      "gracePeriods": 933,
      "thresholdTarget": 283,
      "tolerancePeriod": {
        "startDateTime": "2025-04-01T06:38:38Z",
        "endDateTime": "2025-04-10T06:38:38Z"
      }
    }
  }
}
```


Delete Event

Message example for ThresholdRuleDeleteEvent event


```
Content-Type: application/json
{
  "@type": "ThresholdRuleDeleteEvent",
  "@baseType": "ThresholdRuleDeleteEvent",
  "@schemaLocation": "/schema-registry/ThresholdRuleDeleteEvent.yaml",
  "href": "/tmf-api/performanceManagement/ThresholdRuleDeleteEvent/f3a34f89-0046-
tjb-01",
  "id": "f3a34f89-0046-01",
  "correlationId": "f3a34f89-0046-cr-01",
  "domain": "Commercial",
  "title": "ThresholdRuleDeleteEvent",
  "description": "ThresholdRuleChangeEvent illustration",
  "priority": "5",
  "timeOccurred": "2024-11-18T09:35:14Z",
  "source": {
```


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```
    "@type": "EntityRef",
    "@schemaLocation": "/schema-registry/EntityRef.yaml",
    "href": "/tmf-api/resourceInventoryManagement/v5/logicalResource/f3a34f89-0046-
src-01",
    "id": "f3a34f89-0046-src-01",
    "name": "CRM App",
    "@referredType": "LogicalResource"
  },
  "reportingSystem": {
    "@type": "SoftwareRef",
    "@schemaLocation":
"https://mycsp.com:8080/tmfapi/schema/SoftwareRef/Extended.schema.json",
    "@referredType": "SoftwareResource",
    "id": "533242BC35DSSA55555",
    "href":
"https://mycsp.com:8080/tmfapi/resourceManagement/v5/resource/533242BC35DSSA55555",
    "name": "BlueSkySoftware BSS/OSS"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "@schemaLocation": "/schema-registry/PartyRoleRef.yaml",
        "href": "/tmf-api/customerManagement/v5/customer/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "John Doe",
        "@referredType": "Customer",
        "partyId": "f3a34f89-0046-pt-01",
        "partyName": "John Doe"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/PartyRef.yaml",
        "href": "/tmf-api/partyManagement/v5/organization/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "6d Tech",
        "@referredType": "Organization"
      }
    }
  ],
  "eventId": "f3a34f89-0046-evtid-01",
  "eventTime": "2024-11-18T09:35:14Z",
  "eventType": "ThresholdRuleDeleteEvent",
  "event": {
    "thresholdRule": {
      "@type": "SimpleThresholdRule",
      "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-
01",
      "id": "f3a34f89-0046-thr-01"
    }
  }
}
```


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Attribute Value Change Event

Message example for ThresholdRuleAttributeValueChangeEvent event

```
Content-Type: application/json
{
  "@type": "ThresholdRuleAttributeValueChangeEvent",
  "@baseType": "ThresholdRuleChangeEvent",
  "@schemaLocation": "/schema-registry/ThresholdRuleAttributeValueChangeEvent.yaml",
  "href": "/tmf-
api/performanceManagement/ThresholdRuleAttributeValueChangeEvent/f3a34f89-0046-tjb-
01",
  "id": "f3a34f89-0046-01",
  "correlationId": "f3a34f89-0046-cr-01",
  "domain": "Commercial",
  "title": "ThresholdRuleAttributeValueChangeEvent",
  "description": "ThresholdRuleAttributeValueChangeEvent illustration",
  "priority": "5",
  "timeOccurred": "2024-11-18T09:35:14Z",
  "source": {
    "@type": "EntityRef",
    "@schemaLocation": "/schema-registry/EntityRef.yaml",
    "href": "/tmf-api/resourceInventoryManagement/v5/logicalResource/f3a34f89-0046-
src-01",
    "id": "f3a34f89-0046-src-01",
    "name": "CRM App",
    "@referredType": "LogicalResource"
  },
  "reportingSystem": {
    "@type": "SoftwareRef",
    "@schemaLocation":
"https://mycsp.com:8080/tmfapi/schema/SoftwareRef/Extended.schema.json",
    "@referredType": "SoftwareResource",
    "id": "533242BC35DSSA55555",
    "href":
"https://mycsp.com:8080/tmfapi/resourceManagement/v5/resource/533242BC35DSSA55555",
    "name": "BlueSkySoftware BSS/OSS"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "@schemaLocation": "/schema-registry/PartyRoleRef.yaml",
        "href": "/tmf-api/customerManagement/v5/customer/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "John Doe",
        "@referredType": "Customer",
        "partyId": "f3a34f89-0046-pt-01",
        "partyName": "John Doe"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/PartyRef.yaml",
```

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```
        "href": "/tmf-api/partyManagement/v5/organization/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "6d Tech",
        "@referredType": "Organization"
      }
    }
  ],
  "eventId": "f3a34f89-0046-evtid-01",
  "eventTime": "2024-11-18T09:35:14Z",
  "eventType": "ThresholdRuleAttributeValueChangeEvent",
  "event": {
    "thresholdRule": {
      "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-
01",
      "id": "f3a34f89-0046-thr-01",
      "@type": "SimpleThresholdRule",
      "gracePeriods": 933,
      "thresholdTarget": 283
    }
  }
}
```


ThresholdJob

Create Event

Message example for ThresholdJobCreateEvent event


```
Content-Type: application/json
{
  "@type": "ThresholdJobCreateEvent",
  "@baseType": "ThresholdJobCreateEvent",
  "@schemaLocation": "/schema-registry/ThresholdJobCreateEvent.yaml",
  "href": "/tmf-api/performanceManagement/ThresholdJobCreateEvent/f3a34f89-0046-tjb-
01",
  "id": "f3a34f89-0046-01",
  "correlationId": "f3a34f89-0046-cr-01",
  "domain": "Commercial",
  "title": "ThresholdJobCreateEvent",
  "description": "ThresholdBatchEvent illustration",
  "priority": "5",
  "timeOccurred": "2024-11-18T09:35:14Z",
  "source": {
    "@type": "EntityRef",
    "@schemaLocation": "/schema-registry/EntityRef.yaml",
    "href": "/tmf-api/resourceInventoryManagement/v5/logicalResource/f3a34f89-0046-
src-01",
    "id": "f3a34f89-0046-src-01",
    "name": "CRM App",
    "@referredType": "LogicalResource"
  },
  "reportingSystem": {
    "@type": "SoftwareRef",
    "@schemaLocation":
"https://mycsp.com:8080/tmfapi/schema/SoftwareRef/Extended.schema.json",
    "@referredType": "SoftwareResource",
    "id": "533242BC35DSSA55555",
    "href":
"https://mycsp.com:8080/tmfapi/resourceManagement/v5/resource/533242BC35DSSA55555",
```


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```
      "name": "BlueSkySoftware BSS/OSS"
    },
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
        "role": "initiator",
        "partyOrPartyRole": {
          "@type": "PartyRoleRef",
          "@schemaLocation": "/schema-registry/PartyRoleRef.yaml",
          "href": "/tmf-api/customerManagement/v5/customer/f3a34f89-0046-pt-01",
          "id": "f3a34f89-0046-pt-01",
          "name": "John Doe",
          "@referredType": "Customer",
          "partyId": "f3a34f89-0046-pt-01",
          "partyName": "John Doe"
        }
      },
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
        "role": "initiator",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "@schemaLocation": "/schema-registry/PartyRef.yaml",
          "href": "/tmf-api/partyManagement/v5/organization/f3a34f89-0046-pt-01",
          "id": "f3a34f89-0046-pt-01",
          "name": "6d Tech",
          "@referredType": "Organization"
        }
      }
    ],
    "eventId": "f3a34f89-0046-evtid-01",
    "eventTime": "2024-11-18T09:35:14Z",
    "eventType": "ThresholdJobCreateEvent",
    "event": {
      "thresholdJob": {
        "@type": "ThresholdJob",
        "adminState": "LOCKED",
        "creationDate": "2025-02-12T06:37:22Z",
        "executionState": "Active",
        "granularity": "G_30MN",
        "id": "3a34f89-0046-tjb-01",
        "href": "tmf-api/performanceThresholding/v5/thresholdJob/3a34f89-0046-tjb-01",
        "jobPriority": 610,
        "lastUpdate": "2025-02-12T06:37:22Z",
        "monitoredObjectsCriteria": [
          {
            "monitoredObjectFilter": "nodeName=Router01"
          }
        ],
        "performanceThreshold": {
          "@type": "ThresholdRef",
          "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-pfth-
01",
          "id": "f3a34f89-0046-pfth-01",
          "@referredType": "Threshold"
        },
        "scheduleDefinition": {
          "@type": "ScheduleDefinition",
          "@baseType": "ScheduleDefinition",
          "@schemaLocation": "/schema-registry/v5/ScheduleDefinition.yaml",
          "id": "f3a34f89-0046-sd-01",
```


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```
        "validFor": {
          "startDateTime": "2025-04-01T06:37:22Z",
          "endDateTime": "2025-04-15T06:37:22Z"
        },
        "recurringFrequency": "every 5 minutes",
        "excludedDate": [
          "2025-02-12T06:37:22Z"
        ],
        "scheduleDefinitionHourRange": [
          {
            "startHour": "09:00",
            "endHour": "17:30"
          }
        ],
        "weeklyScheduleDefinition": [],
        "monthlyScheduleDayOfMonthDefinition": [
          "2025-04-01T06:37:22Z",
          "2025-02-15T06:37:22Z"
        ],
        "monthlyScheduleDayOfWeekDefinition": {
          "@type": "MonthlyScheduleDayOfWeekDefinition",
          "@baseType": "MonthlyScheduleDayOfWeekDefinition",
          "@schemaLocation": "/schema-
registry/v5/MonthlyScheduleDayOfWeekDefinition.yaml",
          "dayOfMonthRecurrence": [
            {
              "@type": "DayOfMonthRecurrence",
              "@baseType": "DayOfMonthRecurrence",
              "@schemaLocation": "/schema-registry/v5/DayOfMonthRecurrence.yaml"
            }
          ],
          "recurringDaySequence": "FirstWeekOfMonth"
        }
      },
      "trackingRecord": "Router01_PacketDropRate_ThresholdMonitoring_ExecutionLog"
    }
  }
}
```


Delete Event

Message example for ThresholdJobDeleteEvent event


```
Content-Type: application/json
{
  "@type": "ThresholdJobDeleteEvent",
  "@baseType": "ThresholdJobDeleteEvent",
  "@schemaLocation": "/schema-registry/ThresholdJobDeleteEvent.yaml",
  "href": "/tmf-api/performanceManagement/ThresholdJobDeleteEvent/f3a34f89-0046-tjb-
01",
  "id": "f3a34f89-0046-01",
  "correlationId": "f3a34f89-0046-cr-01",
  "domain": "Commercial",
  "title": "ThresholdJobDeleteEvent",
  "description": "ThresholdBatchEvent illustration",
  "priority": "5",
  "timeOccurred": "2024-11-18T09:35:14Z",
  "source": {
    "@type": "EntityRef",
    "@schemaLocation": "/schema-registry/EntityRef.yaml",
```


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```
    "href": "/tmf-api/resourceInventoryManagement/v5/logicalResource/f3a34f89-0046-
src-01",
    "id": "f3a34f89-0046-src-01",
    "name": "CRM App",
    "@referredType": "LogicalResource"
  },
  "reportingSystem": {
    "@type": "SoftwareRef",
    "@schemaLocation":
"https://mycsp.com:8080/tmfapi/schema/SoftwareRef/Extended.schema.json",
    "@referredType": "SoftwareResource",
    "id": "533242BC35DSSA55555",
    "href":
"https://mycsp.com:8080/tmfapi/resourceManagement/v5/resource/533242BC35DSSA55555",
    "name": "BlueSkySoftware BSS/OSS"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "@schemaLocation": "/schema-registry/PartyRoleRef.yaml",
        "href": "/tmf-api/customerManagement/v5/customer/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "John Doe",
        "@referredType": "Customer",
        "partyId": "f3a34f89-0046-pt-01",
        "partyName": "John Doe"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/PartyRef.yaml",
        "href": "/tmf-api/partyManagement/v5/organization/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "6d Tech",
        "@referredType": "Organization"
      }
    }
  ],
  "eventId": "f3a34f89-0046-evtid-01",
  "eventTime": "2024-11-18T09:35:14Z",
  "eventType": "ThresholdJobDeleteEvent",
  "event": {
    "thresholdJob": {
      "@type": "ThresholdJob",
      "id": "3a34f89-0046-tjb-01",
      "href": "tmf-api/performanceThresholding/v5/thresholdJob/3a34f89-0046-tjb-01"
    }
  }
}
```


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Attribute Value Change Event

Message example for ThresholdJobAttributeValueChangeEvent event

```
Content-Type: application/json
{
  "@type": "ThresholdJobAttributeValueChangeEvent",
  "@baseType": "ThresholdJobAttributeValueChangeEvent",
  "@schemaLocation": "/schema-registry/ThresholdJobAttributeValueChangeEvent.yaml",
  "href": "/tmf-
api/performanceManagement/ThresholdJobAttributeValueChangeEvent/f3a34f89-0046-tjb-
01",
  "id": "f3a34f89-0046-01",
  "correlationId": "f3a34f89-0046-cr-01",
  "domain": "Commercial",
  "title": "ThresholdJobAttributeValueChangeEvent",
  "description": "ThresholdBatchEvent illustration",
  "priority": "5",
  "timeOccurred": "2024-11-18T09:35:14Z",
  "source": {
    "@type": "EntityRef",
    "@schemaLocation": "/schema-registry/EntityRef.yaml",
    "href": "/tmf-api/resourceInventoryManagement/v5/logicalResource/f3a34f89-0046-
src-01",
    "id": "f3a34f89-0046-src-01",
    "name": "CRM App",
    "@referredType": "LogicalResource"
  },
  "reportingSystem": {
    "@type": "SoftwareRef",
    "@schemaLocation":
"https://mycsp.com:8080/tmfapi/schema/SoftwareRef/Extended.schema.json",
    "@referredType": "SoftwareResource",
    "id": "533242BC35DSSA55555",
    "href":
"https://mycsp.com:8080/tmfapi/resourceManagement/v5/resource/533242BC35DSSA55555",
    "name": "BlueSkySoftware BSS/OSS"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "@schemaLocation": "/schema-registry/PartyRoleRef.yaml",
        "href": "/tmf-api/customerManagement/v5/customer/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "John Doe",
        "@referredType": "Customer",
        "partyId": "f3a34f89-0046-pt-01",
        "partyName": "John Doe"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/PartyRef.yaml",
```

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```
        "href": "/tmf-api/partyManagement/v5/organization/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "6d Tech",
        "@referredType": "Organization"
      }
    }
  ],
  "eventId": "f3a34f89-0046-evtid-01",
  "eventTime": "2024-11-18T09:35:14Z",
  "eventType": "ThresholdJobAttributeValueChangeEvent",
  "event": {
    "thresholdJob": {
      "@type": "ThresholdJob",
      "jobPriority": 610,
      "id": "3a34f89-0046-tjb-01",
      "href": "tmf-api/performanceThresholding/v5/thresholdJob/3a34f89-0046-tjb-01"
    }
  }
}
```


State Change Event

Message example for ThresholdJobStateChangeEvent event


```
Content-Type: application/json
{
  "@type": "ThresholdJobStateChangeEvent",
  "@baseType": "ThresholdJobStateChangeEvent",
  "@schemaLocation": "/schema-registry/ThresholdJobStateChangeEvent.yaml",
  "href": "/tmf-api/performanceManagement/ThresholdJobStateChangeEvent/f3a34f89-
0046-tjb-01",
  "id": "f3a34f89-0046-01",
  "correlationId": "f3a34f89-0046-cr-01",
  "domain": "Commercial",
  "title": "ThresholdJobStateChangeEvent",
  "description": "ThresholdJobStateChangeEvent illustration",
  "priority": "5",
  "timeOccurred": "2024-11-18T09:35:14Z",
  "source": {
    "@type": "EntityRef",
    "@schemaLocation": "/schema-registry/EntityRef.yaml",
    "href": "/tmf-api/resourceInventoryManagement/v5/logicalResource/f3a34f89-0046-
src-01",
    "id": "f3a34f89-0046-src-01",
    "name": "CRM App",
    "@referredType": "LogicalResource"
  },
  "reportingSystem": {
    "@type": "SoftwareRef",
    "@schemaLocation":
"https://mycsp.com:8080/tmfapi/schema/SoftwareRef/Extended.schema.json",
    "@referredType": "SoftwareResource",
    "id": "533242BC35DSSA55555",
    "href":
"https://mycsp.com:8080/tmfapi/resourceManagement/v5/resource/533242BC35DSSA55555",
    "name": "BlueSkySoftware BSS/OSS"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
```


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```
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "@schemaLocation": "/schema-registry/PartyRoleRef.yaml",
        "href": "/tmf-api/customerManagement/v5/customer/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "John Doe",
        "@referredType": "Customer",
        "partyId": "f3a34f89-0046-pt-01",
        "partyName": "John Doe"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/PartyRef.yaml",
        "href": "/tmf-api/partyManagement/v5/organization/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "6d Tech",
        "@referredType": "Organization"
      }
    }
  ],
  "eventId": "f3a34f89-0046-evtid-01",
  "eventTime": "2024-11-18T09:35:14Z",
  "eventType": "ThresholdJobStateChangeEvent",
  "event": {
    "thresholdJob": {
      "@type": "ThresholdJob",
      "id": "3a34f89-0046-tjb-01",
      "href": "tmf-api/performanceThresholding/v5/thresholdJob/3a34f89-0046-tjb-01",
      "executionState": "Draft",
      "adminState": "UNLOCKED"
    }
  }
}
```


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API OPERATIONS

Uniform Contract

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and Attribute Selection

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Available Operations

Operations on Threshold

List or find Threshold objects

`GET /threshold?fields=…`

Description

This operation list Threshold entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a list of threshold(s). The given filter criteria is name threshold-1

Request

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```
GET /threshold?name=threshold-1
Content-Type: application/json
```

Response

```
200
[
  {
    "@type": "Threshold",
    "@baseType": "Threshold",
    "@schemaLocation": "/schema-registry/v5/Threshold.yml",
    "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-th-01",
    "id": "f3a34f89-0046-th-01",
    "description": "Network performance threshold for monitoring packet drop rate
and triggering alarms when acceptable limits are exceeded",
    "name": "threshold-1",
    "thresholdRule": [
      {
        "@type": "ThresholdRuleRef",
        "@schemaLocation": "/schema-registry/v5/ThresholdRuleRef.yml",
        "@referredType": "SimpleThresholdRule",
        "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-
01",
        "id": "f3a34f89-0046-thr-01"
      }
    ]
  }
]
```

Retrieves a Threshold by ID

`GET /threshold/{id}?fields=…`

Description

This operation retrieves a Threshold entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a specific threshold with id f3a34f89-0046-th-01.

Request

```
GET /threshold/f3a34f89-0046-th-01
Content-Type: application/json
```

Response

```
200
```

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```
{
  "@type": "Threshold",
  "@baseType": "Threshold",
  "@schemaLocation": "/schema-registry/v5/Threshold.yml",
  "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-th-01",
  "id": "f3a34f89-0046-th-01",
  "description": "Network performance threshold for monitoring packet drop rate and
triggering alarms when acceptable limits are exceeded",
  "name": "threshold-1",
  "thresholdRule": [
    {
      "@type": "ThresholdRuleRef",
      "@schemaLocation": "/schema-registry/v5/ThresholdRuleRef.yml",
      "@referredType": "SimpleThresholdRule",
      "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-
01",
      "id": "f3a34f89-0046-thr-01"
    }
  ]
}
```

Creates a Threshold

`POST /threshold?fields=…`

Description

This operation creates a Threshold entity.

Mandatory Attributes

Mandatory Attributes	Rule
name	
thresholdRule	
@type	M

Usage samples

Creation of a new threshold with POST operation

Request

```
POST /threshold
Content-Type: application/json
{
  "@type": "Threshold",
  "@baseType": "Threshold",
  "@schemaLocation": "/schema-registry/v5/Threshold.yml",
  "description": "Network performance threshold for monitoring packet drop rate and
triggering alarms when acceptable limits are exceeded",
  "name": "threshold-1",
  "thresholdRule": [
    {
      "@type": "ThresholdRuleRef",
      "@schemaLocation": "/schema-registry/v5/ThresholdRuleRef.yml",
```

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```
      "@referredType": "SimpleThresholdRule",
      "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-
01",
      "id": "f3a34f89-0046-thr-01"
    }
  ]
}
```

Response

```
201
{
  "@type": "Threshold",
  "@baseType": "Threshold",
  "@schemaLocation": "/schema-registry/v5/Threshold.yml",
  "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-th-01",
  "id": "f3a34f89-0046-th-01",
  "description": "Network performance threshold for monitoring packet drop rate and
triggering alarms when acceptable limits are exceeded",
  "name": "threshold-1",
  "thresholdRule": [
    {
      "@type": "ThresholdRuleRef",
      "@schemaLocation": "/schema-registry/v5/ThresholdRuleRef.yml",
      "@referredType": "SimpleThresholdRule",
      "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-
01",
      "id": "f3a34f89-0046-thr-01"
    }
  ]
}
```

Updates partially a Threshold

`PATCH /threshold/{id}?fields=…`

Description

This operation allows partial updates of a Threshold entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
description	
href	
id	
name	

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Non Patchable Attributes	Rule
thresholdRule	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule

Usage samples

Here's an example of a request for updating a threshold. This example illustrating patch merge, When PATCH is implemented merge syntax is mandatory

Request

```
PATCH /threshold/f3a34f89-0046-th-01
Content-Type: application/merge-patch+json
{
  "description": "Updated threshold description",
  "@type": "Threshold"
}
```

Response

```
200
{
  "@type": "Threshold",
  "@baseType": "Threshold",
  "@schemaLocation": "/schema-registry/v5/Threshold.yml",
  "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-th-01",
  "id": "f3a34f89-0046-th-01",
  "description": "Updated threshold description",
  "name": "threshold-1",
  "thresholdRule": [
    {
      "@type": "ThresholdRuleRef",
      "@schemaLocation": "/schema-registry/v5/ThresholdRuleRef.yml",
      "@referredType": "SimpleThresholdRule",
      "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-
01",
      "id": "f3a34f89-0046-thr-01"
    }
  ]
}
```

Here's an example of a request for updating a threshold. This example illustrating patch application/json

Request

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```
PATCH /threshold/f3a34f89-0046-th-01
Content-Type: application/json
{
  "description": "Updated threshold description",
  "@type": "Threshold"
}
```

Response

```
200
{
  "@type": "Threshold",
  "@baseType": "Threshold",
  "@schemaLocation": "/schema-registry/v5/Threshold.yml",
  "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-th-01",
  "id": "f3a34f89-0046-th-01",
  "description": "Updated threshold description",
  "name": "threshold-1",
  "thresholdRule": [
    {
      "@type": "ThresholdRuleRef",
      "@schemaLocation": "/schema-registry/v5/ThresholdRuleRef.yml",
      "@referredType": "SimpleThresholdRule",
      "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-
01",
      "id": "f3a34f89-0046-thr-01"
    }
  ]
}
```

Here's an example of a request for updating a threshold. This example illustrating patch json

Request

```
PATCH /threshold/3a34f89-0046-tjb-01
Content-Type: application/json-patch+json
[
  {
    "op": "replace",
    "path": "/description",
    "value": "Updated threshold description"
  }
]
```

Response

```
200
{
  "@type": "Threshold",
  "@baseType": "Threshold",
  "@schemaLocation": "/schema-registry/v5/Threshold.yml",
  "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-th-01",
  "id": "f3a34f89-0046-th-01",
  "description": "Updated threshold description",
```

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```
  "name": "threshold-1",
  "thresholdRule": [
    {
      "@type": "ThresholdRuleRef",
      "@schemaLocation": "/schema-registry/v5/ThresholdRuleRef.yml",
      "@referredType": "SimpleThresholdRule",
      "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-
01",
      "id": "f3a34f89-0046-thr-01"
    }
  ]
}
```

Here's an example of a request for updating a threshold. This example illustrating patch query json

Request

```
PATCH /threshold/f3a34f89-0046-th-01
Content-Type: application/json-patch-query+json
[
  {
    "op": "replace",
    "path": "/thresholdRule",
    "value": [
      {
        "@type": "ThresholdRuleRef",
        "@schemaLocation": "/schema-registry/v5/ThresholdRuleRef.yml",
        "@referredType": "SimpleThresholdRule",
        "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-
02",
        "id": "f3a34f89-0046-thr-02"
      }
    ]
  }
]
```

Response

```
200
{
  "@type": "Threshold",
  "@baseType": "Threshold",
  "@schemaLocation": "/schema-registry/v5/Threshold.yml",
  "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-th-01",
  "id": "f3a34f89-0046-th-01",
  "description": "Network performance threshold for monitoring packet drop rate and
triggering alarms when acceptable limits are exceeded",
  "name": "threshold-1",
  "thresholdRule": [
    {
      "@type": "ThresholdRuleRef",
      "@schemaLocation": "/schema-registry/v5/ThresholdRuleRef.yml",
      "@referredType": "SimpleThresholdRule",
      "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-
02",
      "id": "f3a34f89-0046-thr-02"
```

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```
    }
  ]
}
```

Deletes a Threshold

`DELETE /threshold/{id}`

Description

This operation deletes a Threshold entity.

Usage samples

Here's an example of a request for delete a specific threshold.

Request

```
DELETE /threshold/f3a34f89-0046-th-01
Content-Type: application/json
```

Response

```
204
```

Operations on ThresholdRule

Retrieves a ThresholdRule by ID

`GET /thresholdRule/{id}?fields=…`

Description

This operation retrieves a ThresholdRule entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a specific threshold rule with id f3a34f89-0046-thr-01.

Request

```
GET /thresholdRule/f3a34f89-0046-thr-01
Content-Type: application/json
```

Response

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```
200
{
  "@type": "SimpleThresholdRule",
  "@baseType": "ThresholdRule",
  "@schemaLocation": "/schema-registry/v5/SimpleThresholdRule.yml",
  "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-01",
  "id": "f3a34f89-0046-thr-01",
  "measurement": {
    "@type": "PerformanceMeasurementRef",
    "@schemaLocation": "/schema-registry/v5/PerformanceMeasurementRef.yaml",
    "href": "tmf-api/performanceManagement/v5/performanceMeasurement/f3a34f89-0046-
msr-01",
    "id": "f3a34f89-0046-msr-01",
    "@referredType": "PerformanceMeasurement"
  },
  "consequence": [
    {
      "@type": "Consequence",
      "@baseType": "Consequence",
      "@schemaLocation": "/schema-registry/v5/Consequence.yaml",
      "description": "Raises a major severity alarm when packet drop rate exceeds
the defined threshold of 283 packets per second",
      "name": "Alarm Consequence",
      "prescribedAction": "CreateAlarm"
    }
  ],
  "perfAlarmSpecThresholdCrossingDescription": "Packet drop rate threshold violation
detected - dropped packets per second exceeded acceptable limit",
  "performanceAlarmSpecification": {
    "perfAlarmSpecAdditionalText": "The packet drop rate has exceeded the defined
performance threshold, indicating potential congestion or network instability. This
condition may lead to degraded application performance or increased latency if not
resolved promptly.",
    "perfAlarmSpecAlarmType": "PERFORMANCE_DEGRADATION",
    "perfAlarmSpecSeverity": "MAJOR",
    "perfAlarmSpecSpecificProblem": "Packet drop rate exceeded",
    "perfAlarmProbableCause": "Threshold Crossed"
  },
  "condition": "Raise",
  "name": "DropPacketsHighRule",
  "severity": "MAJOR",
  "conformanceComparatorLower": "GT",
  "conformanceComparatorUpper": "LT",
  "conformancePeriod": {
    "startDateTime": "2025-04-10T06:38:38Z",
    "endDateTime": "2025-04-17T06:38:38Z"
  },
  "conformanceTargetLower": 132,
  "conformanceTargetUpper": 776,
  "gracePeriods": 933,
  "thresholdTarget": 283,
  "tolerancePeriod": {
    "startDateTime": "2025-04-01T06:38:38Z",
    "endDateTime": "2025-04-10T06:38:38Z"
  }
}
```


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List or find ThresholdRule objects

`GET /thresholdRule?fields=…`

Description

This operation list ThresholdRule entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a list of threshold rule(s). The given filter criteria is condition Raise and severity MAJOR

Request

```
GET /thresholdRule?condition=Raise&severity=MAJOR
Content-Type: application/json
```

Response

```
200
[
  {
    "@type": "SimpleThresholdRule",
    "@baseType": "ThresholdRule",
    "@schemaLocation": "/schema-registry/v5/SimpleThresholdRule.yml",
    "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-01",
    "id": "f3a34f89-0046-thr-01",
    "measurement": {
      "@type": "PerformanceMeasurementRef",
      "@schemaLocation": "/schema-registry/v5/PerformanceMeasurementRef.yaml",
      "href": "tmf-api/performanceManagement/v5/performanceMeasurement/f3a34f89-
0046-msr-01",
      "id": "f3a34f89-0046-msr-01",
      "@referredType": "PerformanceMeasurement"
    },
    "consequence": [
      {
        "@type": "Consequence",
        "@baseType": "Consequence",
        "@schemaLocation": "/schema-registry/v5/Consequence.yaml",
        "description": "Raises a major severity alarm when packet drop rate exceeds
the defined threshold of 283 packets per second",
        "name": "Alarm Consequence",
        "prescribedAction": "CreateAlarm"
      }
    ],
    "perfAlarmSpecThresholdCrossingDescription": "Packet drop rate threshold
violation detected - dropped packets per second exceeded acceptable limit",
    "performanceAlarmSpecification": {
      "perfAlarmSpecAdditionalText": "The packet drop rate has exceeded the defined
performance threshold, indicating potential congestion or network instability. This
condition may lead to degraded application performance or increased latency if not
resolved promptly.",
      "perfAlarmSpecAlarmType": "PERFORMANCE_DEGRADATION",
```

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```
      "perfAlarmSpecSeverity": "MAJOR",
      "perfAlarmSpecSpecificProblem": "Packet drop rate exceeded",
      "perfAlarmProbableCause": "Threshold Crossed"
    },
    "condition": "Raise",
    "name": "DropPacketsHighRule",
    "severity": "MAJOR",
    "conformanceComparatorLower": "GT",
    "conformanceComparatorUpper": "LT",
    "conformancePeriod": {
      "startDateTime": "2025-04-10T06:38:38Z",
      "endDateTime": "2025-04-17T06:38:38Z"
    },
    "conformanceTargetLower": 132,
    "conformanceTargetUpper": 776,
    "gracePeriods": 933,
    "thresholdTarget": 283,
    "tolerancePeriod": {
      "startDateTime": "2025-04-01T06:38:38Z",
      "endDateTime": "2025-04-10T06:38:38Z"
    }
  }
]
```


Creates a ThresholdRule

`POST /thresholdRule?fields=…`

Description

This operation creates a ThresholdRule entity.

Mandatory Attributes

Mandatory Attributes	Rule
condition	
name	
severity	
@type	M

Usage samples

Creation of a new threshold rule with POST operation

Request


```
POST /thresholdRule
Content-Type: application/json
{
  "@type": "SimpleThresholdRule",
  "@baseType": "ThresholdRule",
  "@schemaLocation": "/schema-registry/v5/SimpleThresholdRule.yml",
  "measurement": {
    "@type": "PerformanceMeasurementRef",
```


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```
    "@schemaLocation": "/schema-registry/v5/PerformanceMeasurementRef.yaml",
    "href": "tmf-api/performanceManagement/v5/performanceMeasurement/f3a34f89-0046-
msr-01",
    "id": "f3a34f89-0046-msr-01",
    "@referredType": "PerformanceMeasurement"
  },
  "consequence": [
    {
      "@type": "Consequence",
      "@baseType": "Consequence",
      "@schemaLocation": "/schema-registry/v5/Consequence.yaml",
      "description": "Raises a major severity alarm when packet drop rate exceeds
the defined threshold of 283 packets per second",
      "name": "Alarm Consequence",
      "prescribedAction": "CreateAlarm"
    }
  ],
  "perfAlarmSpecThresholdCrossingDescription": "Packet drop rate threshold violation
detected - dropped packets per second exceeded acceptable limit",
  "performanceAlarmSpecification": {
    "perfAlarmSpecAdditionalText": "The packet drop rate has exceeded the defined
performance threshold, indicating potential congestion or network instability. This
condition may lead to degraded application performance or increased latency if not
resolved promptly.",
    "perfAlarmSpecAlarmType": "PERFORMANCE_DEGRADATION",
    "perfAlarmSpecSeverity": "MAJOR",
    "perfAlarmSpecSpecificProblem": "Packet drop rate exceeded",
    "perfAlarmProbableCause": "Threshold Crossed"
  },
  "condition": "Raise",
  "name": "DropPacketsHighRule",
  "severity": "MAJOR",
  "conformanceComparatorLower": "GT",
  "conformanceComparatorUpper": "LT",
  "conformancePeriod": {
    "startDateTime": "2025-04-10T06:38:38Z",
    "endDateTime": "2025-04-17T06:38:38Z"
  },
  "conformanceTargetLower": 132,
  "conformanceTargetUpper": 776,
  "gracePeriods": 933,
  "thresholdTarget": 283,
  "tolerancePeriod": {
    "startDateTime": "2025-04-01T06:38:38Z",
    "endDateTime": "2025-04-10T06:38:38Z"
  }
}
```


Response


```
201
{
  "@type": "SimpleThresholdRule",
  "@baseType": "ThresholdRule",
  "@schemaLocation": "/schema-registry/v5/SimpleThresholdRule.yml",
  "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-01",
  "id": "f3a34f89-0046-thr-01",
  "measurement": {
    "@type": "PerformanceMeasurementRef",
    "@schemaLocation": "/schema-registry/v5/PerformanceMeasurementRef.yaml",
```


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```
    "href": "tmf-api/performanceManagement/v5/performanceMeasurement/f3a34f89-0046-
msr-01",
    "id": "f3a34f89-0046-msr-01",
    "@referredType": "PerformanceMeasurement"
  },
  "consequence": [
    {
      "@type": "Consequence",
      "@baseType": "Consequence",
      "@schemaLocation": "/schema-registry/v5/Consequence.yaml",
      "description": "Raises a major severity alarm when packet drop rate exceeds
the defined threshold of 283 packets per second",
      "name": "Alarm Consequence",
      "prescribedAction": "CreateAlarm"
    }
  ],
  "perfAlarmSpecThresholdCrossingDescription": "Packet drop rate threshold violation
detected - dropped packets per second exceeded acceptable limit",
  "performanceAlarmSpecification": {
    "perfAlarmSpecAdditionalText": "The packet drop rate has exceeded the defined
performance threshold, indicating potential congestion or network instability. This
condition may lead to degraded application performance or increased latency if not
resolved promptly.",
    "perfAlarmSpecAlarmType": "PERFORMANCE_DEGRADATION",
    "perfAlarmSpecSeverity": "MAJOR",
    "perfAlarmSpecSpecificProblem": "Packet drop rate exceeded",
    "perfAlarmProbableCause": "Threshold Crossed"
  },
  "condition": "Raise",
  "name": "DropPacketsHighRule",
  "severity": "MAJOR",
  "conformanceComparatorLower": "GT",
  "conformanceComparatorUpper": "LT",
  "conformancePeriod": {
    "startDateTime": "2025-04-10T06:38:38Z",
    "endDateTime": "2025-04-17T06:38:38Z"
  },
  "conformanceTargetLower": 132,
  "conformanceTargetUpper": 776,
  "gracePeriods": 933,
  "thresholdTarget": 283,
  "tolerancePeriod": {
    "startDateTime": "2025-04-01T06:38:38Z",
    "endDateTime": "2025-04-10T06:38:38Z"
  }
}
```


Updates partially a ThresholdRule

`PATCH /thresholdRule/{id}?fields=…`

Description

This operation allows partial updates of a ThresholdRule entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

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Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
condition	
consequence	
href	
id	
measurement	
name	
perfAlarmSpecThresholdCrossing Description	
performanceAlarmSpecification	
severity	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule

Usage samples

Here's an example of a request for updating a threshold rule. This example illustrating patch json

Request

```
PATCH /thresholdRule/f3a34f89-0046-thr-01
Content-Type: application/json-patch+json
[
  {
    "op": "replace",
    "path": "/perfAlarmSpecThresholdCrossingDescription",
    "value": "Packet drop rate threshold violation detected"
  }
]
```

Response

```
200
{
  "@type": "SimpleThresholdRule",
  "@baseType": "ThresholdRule",
  "@schemaLocation": "/schema-registry/v5/SimpleThresholdRule.yml",
  "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-01",
  "id": "f3a34f89-0046-thr-01",
  "measurement": {
```

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```
    "@type": "PerformanceMeasurementRef",
    "@schemaLocation": "/schema-registry/v5/PerformanceMeasurementRef.yaml",
    "href": "tmf-api/performanceManagement/v5/performanceMeasurement/f3a34f89-0046-
msr-01",
    "id": "f3a34f89-0046-msr-01",
    "@referredType": "PerformanceMeasurement"
  },
  "consequence": [
    {
      "@type": "Consequence",
      "@baseType": "Consequence",
      "@schemaLocation": "/schema-registry/v5/Consequence.yaml",
      "description": "Raises a major severity alarm when packet drop rate exceeds
the defined threshold of 283 packets per second",
      "name": "Alarm Consequence",
      "prescribedAction": "CreateAlarm"
    }
  ],
  "perfAlarmSpecThresholdCrossingDescription": "Packet drop rate threshold violation
detected",
  "performanceAlarmSpecification": {
    "perfAlarmSpecAdditionalText": "The packet drop rate has exceeded the defined
performance threshold, indicating potential congestion or network instability. This
condition may lead to degraded application performance or increased latency if not
resolved promptly.",
    "perfAlarmSpecAlarmType": "PERFORMANCE_DEGRADATION",
    "perfAlarmSpecSeverity": "MAJOR",
    "perfAlarmSpecSpecificProblem": "Packet drop rate exceeded",
    "perfAlarmProbableCause": "Threshold Crossed"
  },
  "condition": "Raise",
  "name": "DropPacketsHighRule",
  "severity": "MAJOR",
  "conformanceComparatorLower": "GT",
  "conformanceComparatorUpper": "LT",
  "conformancePeriod": {
    "startDateTime": "2025-04-10T06:38:38Z",
    "endDateTime": "2025-04-17T06:38:38Z"
  },
  "conformanceTargetLower": 132,
  "conformanceTargetUpper": 776,
  "gracePeriods": 933,
  "thresholdTarget": 283,
  "tolerancePeriod": {
    "startDateTime": "2025-04-01T06:38:38Z",
    "endDateTime": "2025-04-10T06:38:38Z"
  }
}
```


Here's an example of a request for updating a threshold rule. This example illustrating patch query json

Request

```
PATCH /thresholdRule/f3a34f89-0046-thr-01
Content-Type: application/json-patch-query+json
[
  {
    "op": "replace",
    "path": "/consequence/prescribedAction?name=Alarm Consequence",
```

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```
    "value": "cancelAlarm"
  }
]
```

Response

```
200
{
  "@type": "SimpleThresholdRule",
  "@baseType": "ThresholdRule",
  "@schemaLocation": "/schema-registry/v5/SimpleThresholdRule.yml",
  "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-01",
  "id": "f3a34f89-0046-thr-01",
  "measurement": {
    "@type": "PerformanceMeasurementRef",
    "@schemaLocation": "/schema-registry/v5/PerformanceMeasurementRef.yaml",
    "href": "tmf-api/performanceManagement/v5/performanceMeasurement/f3a34f89-0046-
msr-01",
    "id": "f3a34f89-0046-msr-01",
    "@referredType": "PerformanceMeasurement"
  },
  "consequence": [
    {
      "@type": "Consequence",
      "@baseType": "Consequence",
      "@schemaLocation": "/schema-registry/v5/Consequence.yaml",
      "description": "Raises a major severity alarm when packet drop rate exceeds
the defined threshold of 283 packets per second",
      "name": "Alarm Consequence",
      "prescribedAction": "cancelAlarm"
    }
  ],
  "perfAlarmSpecThresholdCrossingDescription": "Packet drop rate threshold violation
detected - dropped packets per second exceeded acceptable limit",
  "performanceAlarmSpecification": {
    "perfAlarmSpecAdditionalText": "The packet drop rate has exceeded the defined
performance threshold, indicating potential congestion or network instability. This
condition may lead to degraded application performance or increased latency if not
resolved promptly.",
    "perfAlarmSpecAlarmType": "PERFORMANCE_DEGRADATION",
    "perfAlarmSpecSeverity": "MAJOR",
    "perfAlarmSpecSpecificProblem": "Packet drop rate exceeded",
    "perfAlarmProbableCause": "Threshold Crossed"
  },
  "condition": "Raise",
  "name": "DropPacketsHighRule",
  "severity": "MAJOR",
  "conformanceComparatorLower": "GT",
  "conformanceComparatorUpper": "LT",
  "conformancePeriod": {
    "startDateTime": "2025-04-10T06:38:38Z",
    "endDateTime": "2025-04-17T06:38:38Z"
  },
  "conformanceTargetLower": 132,
  "conformanceTargetUpper": 776,
  "gracePeriods": 933,
  "thresholdTarget": 283,
  "tolerancePeriod": {
    "startDateTime": "2025-04-01T06:38:38Z",
    "endDateTime": "2025-04-10T06:38:38Z"
```

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```
  }
}
```

Here's an example of a request for updating a threshold rule. This example illustrating patch merge, When PATCH is implemented merge syntax is mandatory

Request

```
PATCH /thresholdRule/f3a34f89-0046-thr-01
Content-Type: application/merge-patch+json
{
  "perfAlarmSpecThresholdCrossingDescription": "Packet drop rate threshold violation
detected",
  "@type": "SimpleThresholdRule"
}
```

Response

```
200
{
  "@type": "SimpleThresholdRule",
  "@baseType": "ThresholdRule",
  "@schemaLocation": "/schema-registry/v5/SimpleThresholdRule.yml",
  "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-01",
  "id": "f3a34f89-0046-thr-01",
  "measurement": {
    "@type": "PerformanceMeasurementRef",
    "@schemaLocation": "/schema-registry/v5/PerformanceMeasurementRef.yaml",
    "href": "tmf-api/performanceManagement/v5/performanceMeasurement/f3a34f89-0046-
msr-01",
    "id": "f3a34f89-0046-msr-01",
    "@referredType": "PerformanceMeasurement"
  },
  "consequence": [
    {
      "@type": "Consequence",
      "@baseType": "Consequence",
      "@schemaLocation": "/schema-registry/v5/Consequence.yaml",
      "description": "Raises a major severity alarm when packet drop rate exceeds
the defined threshold of 283 packets per second",
      "name": "Alarm Consequence",
      "prescribedAction": "CreateAlarm"
    }
  ],
  "perfAlarmSpecThresholdCrossingDescription": "Packet drop rate threshold violation
detected",
  "performanceAlarmSpecification": {
    "perfAlarmSpecAdditionalText": "The packet drop rate has exceeded the defined
performance threshold, indicating potential congestion or network instability. This
condition may lead to degraded application performance or increased latency if not
resolved promptly.",
    "perfAlarmSpecAlarmType": "PERFORMANCE_DEGRADATION",
    "perfAlarmSpecSeverity": "MAJOR",
    "perfAlarmSpecSpecificProblem": "Packet drop rate exceeded",
    "perfAlarmProbableCause": "Threshold Crossed"
  },
  "condition": "Raise",
```

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```
  "name": "DropPacketsHighRule",
  "severity": "MAJOR",
  "conformanceComparatorLower": "GT",
  "conformanceComparatorUpper": "LT",
  "conformancePeriod": {
    "startDateTime": "2025-04-10T06:38:38Z",
    "endDateTime": "2025-04-17T06:38:38Z"
  },
  "conformanceTargetLower": 132,
  "conformanceTargetUpper": 776,
  "gracePeriods": 933,
  "thresholdTarget": 283,
  "tolerancePeriod": {
    "startDateTime": "2025-04-01T06:38:38Z",
    "endDateTime": "2025-04-10T06:38:38Z"
  }
}
```


Here's an example of a request for updating a threshold rule. This example illustrating patch application/json

Request

```
PATCH /thresholdRule/f3a34f89-0046-thr-01
Content-Type: application/json
{
  "perfAlarmSpecThresholdCrossingDescription": "Packet drop rate threshold violation
detected - dropped packets per second exceeded acceptable limit",
  "@type": "SimpleThresholdRule"
}
```

Response


```
200
{
  "@type": "SimpleThresholdRule",
  "@baseType": "ThresholdRule",
  "@schemaLocation": "/schema-registry/v5/SimpleThresholdRule.yml",
  "href": "tmf-api/performanceThresholding/v5/thresholdRule/f3a34f89-0046-thr-01",
  "id": "f3a34f89-0046-thr-01",
  "measurement": {
    "@type": "PerformanceMeasurementRef",
    "@schemaLocation": "/schema-registry/v5/PerformanceMeasurementRef.yaml",
    "href": "tmf-api/performanceManagement/v5/performanceMeasurement/f3a34f89-0046-
msr-01",
    "id": "f3a34f89-0046-msr-01",
    "@referredType": "PerformanceMeasurement"
  },
  "consequence": [
    {
      "@type": "Consequence",
      "@baseType": "Consequence",
      "@schemaLocation": "/schema-registry/v5/Consequence.yaml",
      "description": "Raises a major severity alarm when packet drop rate exceeds
the defined threshold of 283 packets per second",
      "name": "Alarm Consequence",
      "prescribedAction": "CreateAlarm"
```


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```
    }
  ],
  "perfAlarmSpecThresholdCrossingDescription": "Packet drop rate threshold violation
detected - dropped packets per second exceeded acceptable limit",
  "performanceAlarmSpecification": {
    "perfAlarmSpecAdditionalText": "The packet drop rate has exceeded the defined
performance threshold, indicating potential congestion or network instability. This
condition may lead to degraded application performance or increased latency if not
resolved promptly.",
    "perfAlarmSpecAlarmType": "PERFORMANCE_DEGRADATION",
    "perfAlarmSpecSeverity": "MAJOR",
    "perfAlarmSpecSpecificProblem": "Packet drop rate exceeded",
    "perfAlarmProbableCause": "Threshold Crossed"
  },
  "condition": "Raise",
  "name": "DropPacketsHighRule",
  "severity": "MAJOR",
  "conformanceComparatorLower": "GT",
  "conformanceComparatorUpper": "LT",
  "conformancePeriod": {
    "startDateTime": "2025-04-10T06:38:38Z",
    "endDateTime": "2025-04-17T06:38:38Z"
  },
  "conformanceTargetLower": 132,
  "conformanceTargetUpper": 776,
  "gracePeriods": 933,
  "thresholdTarget": 283,
  "tolerancePeriod": {
    "startDateTime": "2025-04-01T06:38:38Z",
    "endDateTime": "2025-04-10T06:38:38Z"
  }
}
```


Deletes a ThresholdRule

`DELETE /thresholdRule/{id}`

Description

This operation deletes a ThresholdRule entity.

Usage samples

Here's an example of a request for delete a specific threshold rule.

Request

```
DELETE /thresholdRule/f3a34f89-0046-thr-01
Content-Type: application/json
```

Response

```
204
```

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Operations on ThresholdJob

List or find ThresholdJob objects

GET /thresholdJob?fields=…

Description

This operation list ThresholdJob entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a list of threshold job(s). The given filter criteria is adminState LOCKED and executionState Active

Request

```
GET /thresholdJob?adminState=LOCKED&executionState=Active
Content-Type: application/json
```

Response

```
200
[
  {
    "@type": "ThresholdJob",
    "adminState": "LOCKED",
    "creationDate": "2025-02-12T06:37:22Z",
    "executionState": "Active",
    "granularity": "G_30MN",
    "id": "3a34f89-0046-tjb-01",
    "href": "tmf-api/performanceThresholding/v5/thresholdJob/3a34f89-0046-tjb-01",
    "jobPriority": 610,
    "lastUpdate": "2025-02-12T06:37:22Z",
    "monitoredObjectsCriteria": [
      {
        "monitoredObjectFilter": "nodeName=Router01"
      }
    ],
    "performanceThreshold": {
      "@type": "ThresholdRef",
      "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-pfth-01",
      "id": "f3a34f89-0046-pfth-01",
      "@referredType": "Threshold"
    },
    "scheduleDefinition": {
      "@type": "ScheduleDefinition",
      "@baseType": "ScheduleDefinition",
      "@schemaLocation": "/schema-registry/v5/ScheduleDefinition.yaml",
      "id": "f3a34f89-0046-sd-01",
      "validFor": {
        "startDateTime": "2025-04-01T06:37:22Z",
        "endDateTime": "2025-04-15T06:37:22Z"
      },
```

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```
      "recurringFrequency": "every 5 minutes",
      "excludedDate": [
        "2025-02-12T06:37:22Z"
      ],
      "scheduleDefinitionHourRange": [
        {
          "startHour": "09:00",
          "endHour": "17:30"
        }
      ],
      "weeklyScheduleDefinition": [],
      "monthlyScheduleDayOfMonthDefinition": [
        "2025-04-01T06:37:22Z",
        "2025-02-15T06:37:22Z"
      ],
      "monthlyScheduleDayOfWeekDefinition": {
        "@type": "MonthlyScheduleDayOfWeekDefinition",
        "@baseType": "MonthlyScheduleDayOfWeekDefinition",
        "@schemaLocation": "/schema-
registry/v5/MonthlyScheduleDayOfWeekDefinition.yaml",
        "dayOfMonthRecurrence": [
          {
            "@type": "DayOfMonthRecurrence",
            "@baseType": "DayOfMonthRecurrence",
            "@schemaLocation": "/schema-registry/v5/DayOfMonthRecurrence.yaml"
          }
        ],
        "recurringDaySequence": "FirstWeekOfMonth"
      }
    },
    "trackingRecord": "Router01_PacketDropRate_ThresholdMonitoring_ExecutionLog"
  }
]
```


Retrieves a ThresholdJob by ID

`GET /thresholdJob/{id}?fields=…`

Description

This operation retrieves a ThresholdJob entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a specific threshold job with id 3a34f89-0046-tjb-01.

Request

```
GET /thresholdJob/3a34f89-0046-tjb-01
Content-Type: application/json
```

Response

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```
200
{
  "@type": "ThresholdJob",
  "adminState": "LOCKED",
  "creationDate": "2025-02-12T06:37:22Z",
  "executionState": "Active",
  "granularity": "G_30MN",
  "id": "3a34f89-0046-tjb-01",
  "href": "tmf-api/performanceThresholding/v5/thresholdJob/3a34f89-0046-tjb-01",
  "jobPriority": 610,
  "lastUpdate": "2025-02-12T06:37:22Z",
  "monitoredObjectsCriteria": [
    {
      "monitoredObjectFilter": "nodeName=Router01"
    }
  ],
  "performanceThreshold": {
    "@type": "ThresholdRef",
    "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-pfth-01",
    "id": "f3a34f89-0046-pfth-01",
    "@referredType": "Threshold"
  },
  "scheduleDefinition": {
    "@type": "ScheduleDefinition",
    "@baseType": "ScheduleDefinition",
    "@schemaLocation": "/schema-registry/v5/ScheduleDefinition.yaml",
    "id": "f3a34f89-0046-sd-01",
    "validFor": {
      "startDateTime": "2025-04-01T06:37:22Z",
      "endDateTime": "2025-04-15T06:37:22Z"
    },
    "recurringFrequency": "every 5 minutes",
    "excludedDate": [
      "2025-02-12T06:37:22Z"
    ],
    "scheduleDefinitionHourRange": [
      {
        "startHour": "09:00",
        "endHour": "17:30"
      }
    ],
    "weeklyScheduleDefinition": [],
    "monthlyScheduleDayOfMonthDefinition": [
      "2025-04-01T06:37:22Z",
      "2025-02-15T06:37:22Z"
    ],
    "monthlyScheduleDayOfWeekDefinition": {
      "@type": "MonthlyScheduleDayOfWeekDefinition",
      "@baseType": "MonthlyScheduleDayOfWeekDefinition",
      "@schemaLocation": "/schema-
registry/v5/MonthlyScheduleDayOfWeekDefinition.yaml",
      "dayOfMonthRecurrence": [
        {
          "@type": "DayOfMonthRecurrence",
          "@baseType": "DayOfMonthRecurrence",
          "@schemaLocation": "/schema-registry/v5/DayOfMonthRecurrence.yaml"
        }
      ],
      "recurringDaySequence": "FirstWeekOfMonth"
    }
  },
  "trackingRecord": "Router01_PacketDropRate_ThresholdMonitoring_ExecutionLog"
```


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```
}
```

Creates a ThresholdJob

POST /thresholdJob?fields=…

Description

This operation creates a ThresholdJob entity.

Mandatory Attributes

Mandatory Attributes	Rule
granularity	
performanceThreshold.href	
@type	M

Usage samples

Creation of a new threshold job with POST operation

Request

```
POST /thresholdJob
Content-Type: application/json
{
  "@type": "ThresholdJob",
  "adminState": "LOCKED",
  "creationDate": "2025-02-12T06:37:22Z",
  "executionState": "Active",
  "granularity": "G_30MN",
  "jobPriority": 610,
  "lastUpdate": "2025-02-12T06:37:22Z",
  "monitoredObjectsCriteria": [
    {
      "monitoredObjectFilter": "nodeName=Router01"
    }
  ],
  "performanceThreshold": {
    "@type": "ThresholdRef",
    "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-pfth-01",
    "id": "f3a34f89-0046-pfth-01",
    "@referredType": "Threshold"
  },
  "scheduleDefinition": {
    "@type": "ScheduleDefinition",
    "@baseType": "ScheduleDefinition",
    "@schemaLocation": "/schema-registry/v5/ScheduleDefinition.yaml",
    "id": "f3a34f89-0046-sd-01",
    "validFor": {
      "startDateTime": "2025-04-01T06:37:22Z",
      "endDateTime": "2025-04-15T06:37:22Z"
    },
    "recurringFrequency": "every 5 minutes",
    "excludedDate": [
```

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```
      "2025-02-12T06:37:22Z"
    ],
    "scheduleDefinitionHourRange": [
      {
        "startHour": "09:00",
        "endHour": "17:30"
      }
    ],
    "weeklyScheduleDefinition": [],
    "monthlyScheduleDayOfMonthDefinition": [
      "2025-04-01T06:37:22Z",
      "2025-02-15T06:37:22Z"
    ],
    "monthlyScheduleDayOfWeekDefinition": {
      "@type": "MonthlyScheduleDayOfWeekDefinition",
      "@baseType": "MonthlyScheduleDayOfWeekDefinition",
      "@schemaLocation": "/schema-
registry/v5/MonthlyScheduleDayOfWeekDefinition.yaml",
      "dayOfMonthRecurrence": [
        {
          "@type": "DayOfMonthRecurrence",
          "@baseType": "DayOfMonthRecurrence",
          "@schemaLocation": "/schema-registry/v5/DayOfMonthRecurrence.yaml"
        }
      ],
      "recurringDaySequence": "FirstWeekOfMonth"
    }
  },
  "trackingRecord": "Router01_PacketDropRate_ThresholdMonitoring_ExecutionLog"
}
```


Response


```
201
{
  "@type": "ThresholdJob",
  "adminState": "LOCKED",
  "creationDate": "2025-02-12T06:37:22Z",
  "executionState": "Active",
  "granularity": "G_30MN",
  "id": "3a34f89-0046-tjb-01",
  "href": "tmf-api/performanceThresholding/v5/thresholdJob/3a34f89-0046-tjb-01",
  "jobPriority": 610,
  "lastUpdate": "2025-02-12T06:37:22Z",
  "monitoredObjectsCriteria": [
    {
      "monitoredObjectFilter": "nodeName=Router01"
    }
  ],
  "performanceThreshold": {
    "@type": "ThresholdRef",
    "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-pfth-01",
    "id": "f3a34f89-0046-pfth-01",
    "@referredType": "Threshold"
  },
  "scheduleDefinition": {
    "@type": "ScheduleDefinition",
    "@baseType": "ScheduleDefinition",
    "@schemaLocation": "/schema-registry/v5/ScheduleDefinition.yaml",
    "id": "f3a34f89-0046-sd-01",
```


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```
    "validFor": {
      "startDateTime": "2025-04-01T06:37:22Z",
      "endDateTime": "2025-04-15T06:37:22Z"
    },
    "recurringFrequency": "every 5 minutes",
    "excludedDate": [
      "2025-02-12T06:37:22Z"
    ],
    "scheduleDefinitionHourRange": [
      {
        "startHour": "09:00",
        "endHour": "17:30"
      }
    ],
    "weeklyScheduleDefinition": [],
    "monthlyScheduleDayOfMonthDefinition": [
      "2025-04-01T06:37:22Z",
      "2025-02-15T06:37:22Z"
    ],
    "monthlyScheduleDayOfWeekDefinition": {
      "@type": "MonthlyScheduleDayOfWeekDefinition",
      "@baseType": "MonthlyScheduleDayOfWeekDefinition",
      "@schemaLocation": "/schema-
registry/v5/MonthlyScheduleDayOfWeekDefinition.yaml",
      "dayOfMonthRecurrence": [
        {
          "@type": "DayOfMonthRecurrence",
          "@baseType": "DayOfMonthRecurrence",
          "@schemaLocation": "/schema-registry/v5/DayOfMonthRecurrence.yaml"
        }
      ],
      "recurringDaySequence": "FirstWeekOfMonth"
    }
  },
  "trackingRecord": "Router01_PacketDropRate_ThresholdMonitoring_ExecutionLog"
}
```


Updates partially a ThresholdJob

`PATCH /thresholdJob/{id}?fields=…`

Description

This operation allows partial updates of a ThresholdJob entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	

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Non Patchable Attributes	Rule
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
adminState	
creationDate	
executionState	
granularity	
jobPriority	
lastUpdate	
monitoredObjectsCriteria	
performanceThreshold	
scheduleDefinition	
trackingRecord	

Usage samples

Here's an example of a request for updating a threshold job. This example illustrating patch query json

Request

```
PATCH /thresholdJob/3a34f89-0046-tjb-01
Content-Type: application/json-patch-query+json
[
  {
    "op": "replace",
    "path":
"/monitoredObjectsCriteria[?(@.monitoredObjectFilter=='nodeName=Router01')]/monitore
dObjectFilter",
    "value": "nodeName=Router02"
  }
]
```

Response

```
200
{
  "@type": "ThresholdJob",
  "adminState": "LOCKED",
  "creationDate": "2025-02-12T06:37:22Z",
  "executionState": "Active",
  "granularity": "G_30MN",
  "id": "3a34f89-0046-tjb-01",
```

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```
  "href": "tmf-api/performanceThresholding/v5/thresholdJob/3a34f89-0046-tjb-01",
  "jobPriority": 610,
  "lastUpdate": "2025-02-12T06:37:22Z",
  "monitoredObjectsCriteria": [
    {
      "monitoredObjectFilter": "nodeName=Router02"
    }
  ],
  "performanceThreshold": {
    "@type": "ThresholdRef",
    "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-pfth-01",
    "id": "f3a34f89-0046-pfth-01",
    "@referredType": "Threshold"
  },
  "scheduleDefinition": {
    "@type": "ScheduleDefinition",
    "@baseType": "ScheduleDefinition",
    "@schemaLocation": "/schema-registry/v5/ScheduleDefinition.yaml",
    "id": "f3a34f89-0046-sd-01",
    "validFor": {
      "startDateTime": "2025-04-01T06:37:22Z",
      "endDateTime": "2025-04-15T06:37:22Z"
    },
    "recurringFrequency": "every 5 minutes",
    "excludedDate": [
      "2025-02-12T06:37:22Z"
    ],
    "scheduleDefinitionHourRange": [
      {
        "startHour": "09:00",
        "endHour": "17:30"
      }
    ],
    "weeklyScheduleDefinition": [],
    "monthlyScheduleDayOfMonthDefinition": [
      "2025-04-01T06:37:22Z",
      "2025-02-15T06:37:22Z"
    ],
    "monthlyScheduleDayOfWeekDefinition": {
      "@type": "MonthlyScheduleDayOfWeekDefinition",
      "@baseType": "MonthlyScheduleDayOfWeekDefinition",
      "@schemaLocation": "/schema-
registry/v5/MonthlyScheduleDayOfWeekDefinition.yaml",
      "dayOfMonthRecurrence": [
        {
          "@type": "DayOfMonthRecurrence",
          "@baseType": "DayOfMonthRecurrence",
          "@schemaLocation": "/schema-registry/v5/DayOfMonthRecurrence.yaml"
        }
      ],
      "recurringDaySequence": "FirstWeekOfMonth"
    }
  },
  "trackingRecord": "Router01_PacketDropRate_ThresholdMonitoring_ExecutionLog"
}
```


Here's an example of a request for updating a threshold job. This example illustrating patch json

Request

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```
PATCH /thresholdJob/3a34f89-0046-tjb-01
Content-Type: application/json-patch+json
[
  {
    "op": "replace",
    "path": "/jobPriority",
    "value": "611"
  }
]
```

Response

```
200
{
  "@type": "ThresholdJob",
  "adminState": "LOCKED",
  "creationDate": "2025-02-12T06:37:22Z",
  "executionState": "Deactive",
  "granularity": "G_30MN",
  "id": "3a34f89-0046-tjb-01",
  "href": "tmf-api/performanceThresholding/v5/thresholdJob/3a34f89-0046-tjb-01",
  "jobPriority": 611,
  "lastUpdate": "2025-02-12T06:37:22Z",
  "monitoredObjectsCriteria": [
    {
      "monitoredObjectFilter": "nodeName=Router01"
    }
  ],
  "performanceThreshold": {
    "@type": "ThresholdRef",
    "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-pfth-01",
    "id": "f3a34f89-0046-pfth-01",
    "@referredType": "Threshold"
  },
  "scheduleDefinition": {
    "@type": "ScheduleDefinition",
    "@baseType": "ScheduleDefinition",
    "@schemaLocation": "/schema-registry/v5/ScheduleDefinition.yaml",
    "id": "f3a34f89-0046-sd-01",
    "validFor": {
      "startDateTime": "2025-04-01T06:37:22Z",
      "endDateTime": "2025-04-15T06:37:22Z"
    },
    "recurringFrequency": "every 5 minutes",
    "excludedDate": [
      "2025-02-12T06:37:22Z"
    ],
    "scheduleDefinitionHourRange": [
      {
        "startHour": "09:00",
        "endHour": "17:30"
      }
    ],
    "weeklyScheduleDefinition": [],
    "monthlyScheduleDayOfMonthDefinition": [
      "2025-04-01T06:37:22Z",
      "2025-02-15T06:37:22Z"
    ],
    "monthlyScheduleDayOfWeekDefinition": {
```

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```
      "@type": "MonthlyScheduleDayOfWeekDefinition",
      "@baseType": "MonthlyScheduleDayOfWeekDefinition",
      "@schemaLocation": "/schema-
registry/v5/MonthlyScheduleDayOfWeekDefinition.yaml",
      "dayOfMonthRecurrence": [
        {
          "@type": "DayOfMonthRecurrence",
          "@baseType": "DayOfMonthRecurrence",
          "@schemaLocation": "/schema-registry/v5/DayOfMonthRecurrence.yaml"
        }
      ],
      "recurringDaySequence": "FirstWeekOfMonth"
    }
  },
  "trackingRecord": "Router01_PacketDropRate_ThresholdMonitoring_ExecutionLog"
}
```


Here's an example of a request for updating a threshold job. This example illustrating patch merge, When PATCH is implemented merge syntax is mandatory

Request

```
PATCH /thresholdJob/3a34f89-0046-tjb-01
Content-Type: application/merge-patch+json
{
  "jobPriority": 611,
  "@type": "ThresholdJob"
}
```

Response


```
200
{
  "@type": "ThresholdJob",
  "adminState": "LOCKED",
  "creationDate": "2025-02-12T06:37:22Z",
  "executionState": "Active",
  "granularity": "G_30MN",
  "id": "3a34f89-0046-tjb-01",
  "href": "tmf-api/performanceThresholding/v5/thresholdJob/3a34f89-0046-tjb-01",
  "jobPriority": 611,
  "lastUpdate": "2025-02-12T06:37:22Z",
  "monitoredObjectsCriteria": [
    {
      "monitoredObjectFilter": "nodeName=Router01"
    }
  ],
  "performanceThreshold": {
    "@type": "ThresholdRef",
    "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-pfth-01",
    "id": "f3a34f89-0046-pfth-01",
    "@referredType": "Threshold"
  },
  "scheduleDefinition": {
    "@type": "ScheduleDefinition",
    "@baseType": "ScheduleDefinition",
    "@schemaLocation": "/schema-registry/v5/ScheduleDefinition.yaml",
```


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```
    "id": "f3a34f89-0046-sd-01",
    "validFor": {
      "startDateTime": "2025-04-01T06:37:22Z",
      "endDateTime": "2025-04-15T06:37:22Z"
    },
    "recurringFrequency": "every 5 minutes",
    "excludedDate": [
      "2025-02-12T06:37:22Z"
    ],
    "scheduleDefinitionHourRange": [
      {
        "startHour": "09:00",
        "endHour": "17:30"
      }
    ],
    "weeklyScheduleDefinition": [],
    "monthlyScheduleDayOfMonthDefinition": [
      "2025-04-01T06:37:22Z",
      "2025-02-15T06:37:22Z"
    ],
    "monthlyScheduleDayOfWeekDefinition": {
      "@type": "MonthlyScheduleDayOfWeekDefinition",
      "@baseType": "MonthlyScheduleDayOfWeekDefinition",
      "@schemaLocation": "/schema-
registry/v5/MonthlyScheduleDayOfWeekDefinition.yaml",
      "dayOfMonthRecurrence": [
        {
          "@type": "DayOfMonthRecurrence",
          "@baseType": "DayOfMonthRecurrence",
          "@schemaLocation": "/schema-registry/v5/DayOfMonthRecurrence.yaml"
        }
      ],
      "recurringDaySequence": "FirstWeekOfMonth"
    }
  },
  "trackingRecord": "Router01_PacketDropRate_ThresholdMonitoring_ExecutionLog"
}
```


Here's an example of a request for updating a threshold job. This example illustrating patch application/json

Request

```
PATCH /thresholdJob/3a34f89-0046-tjb-01
Content-Type: application/json
{
  "jobPriority": 611,
  "@type": "ThresholdJob"
}
```

Response


```
200
{
  "@type": "ThresholdJob",
  "adminState": "LOCKED",
  "creationDate": "2025-02-12T06:37:22Z",
```


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```
  "executionState": "Active",
  "granularity": "G_30MN",
  "id": "3a34f89-0046-tjb-01",
  "href": "tmf-api/performanceThresholding/v5/thresholdJob/3a34f89-0046-tjb-01",
  "jobPriority": 611,
  "lastUpdate": "2025-02-12T06:37:22Z",
  "monitoredObjectsCriteria": [
    {
      "monitoredObjectFilter": "nodeName=Router01"
    }
  ],
  "performanceThreshold": {
    "@type": "ThresholdRef",
    "href": "tmf-api/performanceThresholding/v5/threshold/f3a34f89-0046-pfth-01",
    "id": "f3a34f89-0046-pfth-01",
    "@referredType": "Threshold"
  },
  "scheduleDefinition": {
    "@type": "ScheduleDefinition",
    "@baseType": "ScheduleDefinition",
    "@schemaLocation": "/schema-registry/v5/ScheduleDefinition.yaml",
    "id": "f3a34f89-0046-sd-01",
    "validFor": {
      "startDateTime": "2025-04-01T06:37:22Z",
      "endDateTime": "2025-04-15T06:37:22Z"
    },
    "recurringFrequency": "every 5 minutes",
    "excludedDate": [
      "2025-02-12T06:37:22Z"
    ],
    "scheduleDefinitionHourRange": [
      {
        "startHour": "09:00",
        "endHour": "17:30"
      }
    ],
    "weeklyScheduleDefinition": [],
    "monthlyScheduleDayOfMonthDefinition": [
      "2025-04-01T06:37:22Z",
      "2025-02-15T06:37:22Z"
    ],
    "monthlyScheduleDayOfWeekDefinition": {
      "@type": "MonthlyScheduleDayOfWeekDefinition",
      "@baseType": "MonthlyScheduleDayOfWeekDefinition",
      "@schemaLocation": "/schema-
registry/v5/MonthlyScheduleDayOfWeekDefinition.yaml",
      "dayOfMonthRecurrence": [
        {
          "@type": "DayOfMonthRecurrence",
          "@baseType": "DayOfMonthRecurrence",
          "@schemaLocation": "/schema-registry/v5/DayOfMonthRecurrence.yaml"
        }
      ],
      "recurringDaySequence": "FirstWeekOfMonth"
    }
  },
  "trackingRecord": "Router01_PacketDropRate_ThresholdMonitoring_ExecutionLog"
}
```


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Deletes a ThresholdJob

DELETE /thresholdJob/{id}

Description

This operation deletes a ThresholdJob entity.

Usage samples

Here's an example of a request for delete a specific threshold job.

Request

```
DELETE /thresholdJob/f3a34f89-0046-thr-01
Content-Type: application/json
```

Response

```
204
```

Operations on AlgorithmThresholdRule

See operations on ThresholdRule.

Operations on SimpleThresholdRule

See operations on ThresholdRule.

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description

Contributors to Document

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